

Revealing Borrowers' Information Through Credit Scores

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May 20, 2026

Abstract

This paper examines how expanding the information content of credit scores affects individuals' access to credit, default rates, and entrepreneurial activity. We analyze a nationwide reform in Brazil that allowed individuals' payment histories to be incorporated into credit scores that previously relied only on financial delinquency records. We partnered with the country's largest credit bureau to access rich microdata, including credit scores constructed under both the old and new information regimes for the same individuals. We combine a model of lenders' decision-making under imperfect information, characterizing how credit allocation shifts across the joint distribution of old and new credit scores, with a series of difference-in-differences designs that causally estimate the policy's effects. We separately identify the effects driven by changes in score values from the score precision, both of which contribute to the total effect. Overall, the policy reduced default rates by shifting credit towards less-risky borrowers, but it also increased credit inequality, negatively affecting nonwhite and less-educated individuals. Looking at effects beyond credit markets, we show that while the reform did not affect the rate of firm creation, it improved firm survival among entrepreneurs it benefited.

JEL Codes: G28, G21, D14

*This paper has been conducted in partnership with **DataLab SERASA Experian**. We would like to thank Mariana Mercucci, Josival Leite, Olivier Devaux, Amilton Pimenta, and Sergio Azevedo for their help in the project. The analysis and conclusions set forth here are those of the authors and do not indicate concurrence by other members of the partner institution. This paper also benefited from the guidance of David Card, Fred Finan, Pat Kline, and Gustavo Manso. We want to thank Andrea Cerrato, Ted Miguel, Sebastián Otero, João Francisco Cabral Perez, Demian Pouzo, Will Sandholtz, Damián Vergara, and Chris Walters for the discussions, comments, and suggestions that helped substantially in improving the paper. All remaining errors are ours. Corresponding author Roberto Hsu Rocha, NOVA SBE, e-mail: roberto.hsu.rocha@novasbe.pt.

1 Introduction

Imperfect information can distort the allocation of credit, compromising the overall efficiency of credit markets and economic activity (Stiglitz and Weiss, 1981). In response, countries have developed credit reporting systems that consolidate information on borrowers’ credit histories and provide summary credit scores. Still, countries widely differ in the amount and type of information available to build such scores (Djankov et al., 2007). Empirically studying how information affects credit allocation through credit scores is challenging. Significant reforms to credit reporting systems are rare and often occur in contexts with insufficient data. Even when such reforms can be studied, revealing additional information affects not only the values of credit scores, but also how informative those scores are to lenders, making it difficult to disentangle the different channels through which information affects credit allocation.

In this paper, we study a large and universal expansion in the information available to credit bureaus for building credit scores. In April 2019, Brazil’s Congress transformed the country’s credit reporting system from an opt-in to an opt-out regime. Before the reform, credit bureaus had full access only to records on financial delinquency information, such as loan defaults, late payments, and other negative financial events. They could access positive financial information, such as on-time payment, successful loan repayments, and the length of credit history, only for individuals who opted into the system. After the policy, positive financial information was automatically shared with credit bureaus for virtually the entire adult population. Within two years, more than 100 million individuals had active records in the new system, up from fewer than 5 million under the opt-in regime. In May 2021, credit bureaus began to commercialize the new credit scores with financial institutions.¹

This reform was a major change to the information environment in one of the world’s largest credit markets. We address central questions around this policy. First, we study the effects on credit reallocation. We show that it depends not only on changes in credit scores’ values, but also on how informative those scores are to lenders. We then characterize the winners and losers from the reform and examine the trade-off between inequality and the efficiency of credit allocation. Lastly, we estimate the effects that go beyond the credit market, affecting business creation and firms’ outcomes.

Causally answering these questions faces measurement and identification challenges. On the measurement side, we partner with the largest credit bureau in the country to

¹Since the population under the opt-in system was small and selected, financial institutions reported not to use the credit scores built with positive information for anyone prior to the reform (BACEN, 2021).

access individual-level financial data, which we combine with the universe of firm ownership records. Importantly, our partners provided us with credit scores built under the old and new systems for the same individuals, allowing us to precisely observe the variation caused by the policy. On the identification side, we develop a conceptual framework for lender decision-making under imperfect information that guides our empirical strategy. This framework motivates a series of difference-in-differences estimates that separately identify how changes in score values and score informativeness affect credit allocation.

Our framework highlights two channels through which additional information affects credit allocation. First, borrowers' credit scores may change under the new information regime, directly affecting lenders' assessments of creditworthiness. Second, even when score values remain unchanged, richer information can alter how informative those scores are to lenders, changing how strongly lenders respond to a given score. These mechanisms imply that understanding the effects of information requires studying the joint distribution of credit scores under the old and new systems, rather than focusing only on average changes in score values.

The framework also guides our empirical strategy. It predicts that borrowers whose scores remain unchanged and equal lenders' prior assessments should be unaffected by the reform, providing a natural control group for the difference-in-differences analysis. More broadly, the joint distribution of old and new credit scores provides the appropriate comparisons to separately identify the effects of changes in score values and changes in score informativeness on credit allocation.

We find that the policy substantially altered the allocation of credit, and that both changes in score values and changes in score informativeness contributed to these effects. Changes in score values had large effects on credit allocation. Individuals who experienced a one-standard-deviation increase in credit scores saw their credit increase by about R\$ 15,000, or 20%, two years after the reform. The effects are symmetric, as individuals whose scores declined by one standard deviation experienced similarly large reductions in credit access. Changes in score informativeness also mattered. Individuals with above-average pre-policy scores received around R\$ 5,000 more in credit, while those with below-average scores lost around R\$ 12,000. Estimates across the joint distribution of old and new credit scores show that both mechanisms operate broadly throughout the distribution.

The reallocation of credit had important distributional consequences for two reasons. First, more informative credit scores allow lenders to place greater weight on observable measures of creditworthiness when allocating credit. Second, given the nature of the information being revealed, individuals with greater pre-policy credit access experienced

larger increases in their credit scores on average. As a result, the variance of credit distribution rose by 2.5%, pushing the Lorenz Curve of credit access further from the equality line. When focusing on specific demographics, the policy widened the credit access gap between white and non-white by R\$ 4,600. This reflects a composition effect, nonwhite individuals are disproportionately concentrated in the lower parts of the joint distribution of credit scores. More broadly, our findings suggest that expansions in information can widen racial disparities in credit markets, contrasting with recent evidence in [Blattner and Nelson \(2024\)](#).

We then examine whether expanding the information available to construct credit scores improved the efficiency of credit allocation by reducing the default rates associated with marginal credit. The overall effect depends on two margins: the default rate of the new credit induced by the policy and the default rate of the credit displaced by it. Under the additional assumption that the reform did not affect the default rates of infra-marginal credit, we combine individual-level data on credit and financial delinquency to estimate the default rates associated with policy-induced credit reallocation. We find that the credit extended because of the additional information carried a default rate of about 3%, while the credit it displaced carried a default rate of about 15%. These estimates suggest that the reform reallocated credit toward borrowers with substantially lower ex-post default rates.

Finally, we ask whether the reallocation of credit had consequences beyond the credit market. We focus on entrepreneurial activity, building on the well-established link between access to credit and the decision to start a firm ([Evans and Jovanovic, 1989](#)), which has long been viewed as an important determinant of economic development ([Banerjee and Newman, 1993](#); [Banerjee and Duflo, 2005](#)). To study these effects, we match the credit bureau data to administrative firm records covering the universe of formal firms in São Paulo, Brazil’s most populous state. This setting allows us to examine not only whether improved credit access affected firm creation, but also whether changes in credit allocation altered the composition and performance of new entrepreneurs.

We find no effect of changes in credit access on the probability that an individual starts a business, an informative null consistent with other evidence on the effects of personal credit in developing countries ([Banerjee et al., 2015](#)). However, the reform affected the performance of new firms. Firms created by individuals who gained credit access because of the reform were more likely to survive two years after creation and exhibited higher levels of economic activity. As a result, even though the policy did not affect firm creation rates, it changed the composition of surviving firms by increasing the share of surviving firms associated with individuals positively affected by the expansion

in information. These findings suggest that improvements in information may affect not only the quantity and allocation of credit in the economy, but also the composition of firms that remain active over time.

Related Literature This paper relates to a broad literature studying how information affects decision-making under imperfect information. A large empirical literature examines how changes in the availability of borrower information affect access to credit and borrowing outcomes (Musto, 2004; Bos and Nakamura, 2014; Dobbie et al., 2020; DeFusco et al., 2022; Jansen et al., 2022; Herkenhoff et al., 2021; Gross et al., 2020). Existing work primarily studies settings in which information shocks change the average assessment of creditworthiness for a given group of borrowers. Our paper shows that richer information can also affect credit allocation by changing how informative credit scores are to lenders, even absent changes in average score values. By observing credit scores built under both the old and new information regimes for the same individuals, we are able to separately characterize the effects of changes in score values and score informativeness on credit allocation and default rates.

Our paper also relates to a longstanding literature on statistical discrimination and noisy signals (Aigner and Cain, 1977). Recent empirical work shows that the informativeness of signals affects decision-making in settings such as bail (Arnold et al., 2022), employment (Bartik and Nelson, 2025), medicine (Chan et al., 2022), and credit markets (Blattner and Nelson, 2024). Relative to this literature, our setting provides two important advantages. First, credit scores provide a direct and ordinal measure of the information available to lenders. Second, the reform generates exogenous variation in the information embedded in these scores. Together, these features allow us to map the effects of additional information to the joint distribution of borrower assessments under the old and new credit scoring systems.

More broadly, the paper contributes to the literature on information-sharing institutions and financial development (Pagano and Jappelli, 1993; Jappelli and Pagano, 2002; Djankov et al., 2007). Existing within-country studies typically focus on firms and compare financing outcomes across firms with differential information availability (Hertzberg et al., 2011; Behr and Sonnekalb, 2012). Our results highlight both the distributional and allocative consequences of information-sharing institutions. The reform increased dispersion in credit access and widened racial inequality in credit markets, while simultaneously reallocating credit toward borrowers with substantially lower ex-post default rates.

Finally, we contribute to the literature studying the effects of financial frictions on entrepreneurship and economic development (Evans and Jovanovic, 1989; Black and

Strahan, 2002; Hurst and Lusardi, 2004; McKenzie, 2017). Consistent with recent evidence on personal credit expansions in developing countries (Banerjee et al., 2015), we find no effects on the extensive margin of firm creation. However, improvements in credit allocation affected the survival of new firms, altering the composition of firms that remain active over time. These findings suggest that information-sharing institutions can affect not only the allocation of credit, but also the composition of economic activity enabled by financial markets.

2 Institutional Background and Data

Brazil is the largest country in Latin America and is characterized by a high level of banking penetration relative to its income level. In 2021, over 80% of the population has a bank account, and individuals hold accounts at an average of 5.4 different banks. Figure A1 benchmarks these patterns against other countries by GDP per capita. There are over 150 operating banks in the country and more than 600 additional financial institutions, such as fintechs and credit cooperatives, that also provide loans.

Credit bureaus have historically been the main institutions responsible for assessing creditworthiness in Brazil. Operating since at least the 1950s and formally regulated by Law 8.078/1990 (the *Código de Defesa do Consumidor*), they relied primarily on a delinquency registry, known as the *Cadastro Negativo de Devedores*, as their main source of information until 2019. This registry tracked defaults, protests, bounced checks, overdue debts, and other financial irregularities. These complaints could be made by creditor companies, such as banks, financial institutions, stores, and service providers. Since 1990, the inclusion of a consumer’s name on defaulters’ lists must comply with federal guidelines, ensuring the consumer’s right to information and to challenge. On top of delinquency records, credit bureaus also used publicly available information to calculate credit scores. These included prosecution records, firm ownership, and bankruptcy records.

2.1 *Cadastro Positivo* - increase in information available to Credit Bureaus

To expand the information available for credit assessment, the Brazilian government introduced a reform known as *Cadastro Positivo*, which broadened the set of data that credit bureaus could use to construct credit scores. In addition to the previously available sources, credit bureaus gained access to *positive financial data*, including records of on-time payments, responsible credit use, the opening of new credit accounts, successful loan repayments, and the length of an individual’s credit history.

Lenders do not access the registry information directly. The data is shared only with authorized credit bureaus responsible for processing the information and generating credit scores for individuals and businesses. These credit scores are subsequently made available to potential lenders and creditors.²

The Cadastro Positivo was initially established by Law No. 12.414 on June 9, 2011. The first information from Cadastro Positivo became available to credit bureaus in the first half of 2014. However, individuals had to opt in to have their information included. By 2019, less than 5% of the population had opted in. [BACEN \(2021\)](#) documents that, due to the low take-up of the policy, all major banks reported not using credit scores built with Cadastro Positivo information in their lending decisions during the opt-in phase.

Given the low take-up and the de facto ineffectiveness of Cadastro Positivo under the opt-in regime, Congress approved Complementary Law 166/2019 in April 2019, which shifted the system from opt-in to opt-out. Rather than requiring individuals to actively enroll, all individuals and businesses with financial records were automatically included in Cadastro Positivo unless they expressly opted out. In practice, this marked the effective implementation of the policy, expanding coverage from almost no individuals to over 100 million individuals with active records. Even though individuals could still opt out, by the end of 2020 fewer than 350,000 people had done so.

The implementation of the new law unfolded over the following two years. Between April and December 2019, regulatory agencies established the conditions for credit bureaus to access the new data, and by December 2019 individuals who were previously outside Cadastro Positivo were automatically included in the system. During the first half of 2020, financial institutions registered with the Central Bank began sharing information with credit bureaus, including consumers' payment histories from the previous two years. Given the complexity of incorporating this new information into credit assessment, credit scores under the new regime were only commercialized in May 2021, marking the point at which lenders could effectively use the expanded information set.³

²Brazil has authorized four credit bureaus to access and employ Cadastro Positivo data for credit assessment purposes. These official credit bureaus include Serasa Experian, Boa Vista SCPC (Serviço Central de Proteção ao Crédito), Quod, and SPC Brasil (Serviço de Proteção ao Crédito).

³Before *Cadastro Positivo*, Brazil already had a robust and established credit market. Therefore large changes in aggregate credit patterns might not be expected to occur due to the policy. We show some aggregate statistics of household credit in Figure [A2](#).

2.2 Data and Sample Construction

Our primary data source comes from a novel research partnership with SERASA Experian, the largest credit bureau in Brazil, and one of four institutions authorized by the Central Bank to access Cadastro Positivo data. For each individual in our sample, SERASA provided us their credit scores constructed with and without the Cadastro Positivo information, as well as the outstanding balances and delinquency amounts underlying those scores. This allows us to observe both the old and new creditworthiness assessments for the same individual at the same point in time, a key feature of our empirical strategy. We complement this with firm ownership records from São Paulo’s trade board (JUCESP), which covers the universe of formal firms created in the state between 2003 and 2023 and includes each founder’s social security number, allowing us to match firm creation to individual credit records.⁴

Our main sample comprises approximately 200,000 randomly selected individuals aged 20 to 60 in 2019 who resided in São Paulo, roughly 1% of the state’s adult population. We focus on São Paulo due to data availability; it is the most populous state in the country, accounting for around one-third of national economic activity. Credit scores in both systems are normalized to have a mean of zero and a standard deviation of 1. Our main outcome, total credit, is defined as the sum of all outstanding balances across active accounts at a given period, following the definition in Gibbs et al. (2024). Financial delinquency is the amount reported to the credit bureau by financial institutions when an individual fails to meet obligations on time.⁵ Summary statistics are presented in Table 1.

For the entrepreneurship analysis in Section 6, we expand the sample by adding approximately 560,000 randomly selected individuals from the same pool of São Paulo adults. Due to the data agreement, we could not obtain individual-level credit and default information for them (this is why this sample is not included in the other analysis). For these 750,000 individuals, we observe credit scores in both systems and match them to the universe of formal firm ownership records from JUCESP. Table A4 shows that the two samples are nearly identical in demographic and credit score characteristics.

⁴Appendix B provides details on our data agreement, including limitations on which information we can access. It also specifies how we conducted the data collection for JUCESP, including scraping more than 4 million files.

⁵In Appendix B we detail the sample construction procedures, indicating all the steps done by the company’s researchers and de-identification processes that were required in our data agreement.

Table 1: Summary Statistics

Demographic Characteristics		Credit Characteristics		
	(1)		(2)	(3)
	Mean		Mean	Std. Dev.
Female	0.45	C. Sc. - Old System	467.15	310.59
Nonwhite	0.24	C. Sc. - New System	552.89	222.37
Less than H.S.	0.21	Total Credit	30325.63	100943.65
High School	0.56	Outstanding Loans	26870.59	97722.95
Some College	0.22	Credit Purchases	3455.04	9243.21
Age	40.19	Financial System Default	2435.93	95206.02
		Other Default	600.01	8291.51
N.Obs	194247			

This Table presents summary statistics of our sample. Statistics are calculated in the last period before the policy’s implementation. Monetary values are presented in Dec. 2023 Brazilian Reais (1 USD = 5.03 BRL).

3 Conceptual Framework

In this section, we develop a framework for lender decision-making under imperfect information, where the lender observes an unbiased signal of borrowers’ true creditworthiness. Our framework serves two main purposes. First, it characterizes how a policy that increases the informativeness of credit scores affects credit allocation. We show that under simplified assumptions, changes in credit allocation can be directly mapped to the joint distribution of signals under old and new policy regimes. Second, we use our framework to guide our empirical strategy, which is especially important in a setting where everyone’s credit scores were affected.

We show that for a given borrower, the more informative signal can simultaneously perturb two distinct objects: the *value* of the signal and its *precision*. Then, a standard difference-in-differences comparing those who changed their credit scores with those who did not conflates both channels and recovers neither structural parameter cleanly. The framework identifies the correct control group for each exercise, and we show that these choices follow from the structure of the expected creditworthiness given the signal received by the decision maker.

While our question centers on the role of information in credit allocation, our simple framework is generalizable to other types of decision-making under imperfect information. The key distinction from other settings is the combination of (i) an ordinal measure of signals and (ii) a policy shift that allows us to observe ordinal signals under two dif-

ferent states of the world. The ordinality of the signal measure is non-trivial to map our framework into empirical comparisons, as it allows us to identify who are the individuals benefited by the precision channel.

For exposition purposes, we focus on intuitive aspects of the framework, and we are relatively loose on formal assumptions and mathematical derivations. In Appendix C, we develop our conceptual framework under formal assumptions and derivations.

3.1 The Relationship Between Credit and Signals

We consider a lender’s decision problem, in which they face borrowers who vary in their true creditworthiness, θ_i . Instead of observing θ , the lender observes a signal realization s_i and forms a posterior belief about the borrower’s type. Her optimal credit decision solves:⁶

$$C^*(s_i) = \operatorname{argmax}_C \mathbb{E}[\pi(C, \theta) | s_i]$$

where $\pi(C, \theta)$ is the lender’s profit function, increasing in θ for any given level of credit C . We assume that the solution to this problem depends on the signal only through the posterior mean of creditworthiness.⁷

$$C^*(s_i) = g(\mathbb{E}[\theta | s_i]), \quad g'(\cdot) > 0 \quad (1)$$

Equation 1 is the structural credit equation. It says that observed credit outcomes are determined by the lender’s posterior assessment of the borrower’s creditworthiness. The function $g(\cdot)$ captures the mapping from beliefs to actions, reflecting competition, regulatory constraints, etc., but its specific form is not needed for our identification argument.

We are interested in a policy that generates a new, more informative, signal s'_i . Under our simplified assumptions, we can write the policy-induced change in credit for individual i as:

$$\Delta C_i(s_i, s'_i) = C^*(s'_i) - C^*(s_i) = g(\mathbb{E}[\theta | s'_i]) - g(\mathbb{E}[\theta | s_i]) \quad (2)$$

The change in credit for individual i due to the policy is exactly what we aim to estimate. The key observation is that ΔC_i depends on the policy only through the

⁶Signals in our framework are not endogenously determined by borrowers in the spirit of [Spence \(1973\)](#). We think of them as information structures as in [Green and Stokey \(1978\)](#) or more recently [Brooks et al. \(2022a,b\)](#)

⁷Appendix C formalizes the conditions for this assumption.

change in the lender’s posterior mean, which in turn depends on the signals they receive.

3.2 Policy under Gaussian Parametrization

In order to provide more intuition on the effects of the policy, we provide a simple Gaussian Parametrization to the information structure. Appendix C extends the framework beyond the Gaussian parametrization. We assume that an individual i ’s type θ_i is drawn from an underlying distribution $\theta_i \sim N(\mu, \sigma^2)$. Their signal is a noisy measure of their creditworthiness:

$$s_i = \theta_i + u_i, \quad u_i = \epsilon_i - \Delta_i$$

where Δ_i and ϵ_i are independent and distributed according to $N(0, \sigma_\Delta^2)$ and $N(0, \sigma_\epsilon^2)$ respectively. Thus $u_i \sim N(0, \sigma_\Delta^2 + \sigma_\epsilon^2)$. This implies that the lender’s expected value about a given borrower’s type is then given by:

$$\mathbb{E}[\theta|s_i] = \mu + \underbrace{\frac{\sigma^2}{\sigma^2 + \sigma_\Delta^2 + \sigma_\epsilon^2}}_{\equiv \lambda} (s_i - \mu)$$

where the expected value is a linear combination of the prior of the lender, given by the population average, and the signal realization s_i . The *Precision Weight*, λ establishes how much weight the decision maker puts on the signal relative to their prior. It captures how informative the signal is. If it was fully informative ($\sigma_\epsilon^2 + \sigma_\Delta^2 = 0$), the expected value of the individual’s type would be equal to the signal. The only difference between the expression above and standard frameworks is that the variance of the *noise* is determined by $\sigma_u^2 = \sigma_\epsilon^2 + \sigma_\Delta^2$.

The policy. We assume a policy that makes Δ_i observable: lender now observe $s'_i = s_i + \Delta_i = \theta_i + \epsilon_i$, which eliminates the Δ_i component of the noise.

This policy maps the idea of Cadastro Positivo, which incorporates more information into the construction of credit scores. For a given individual, the new signal that is observed by the lender can be higher or lower, depending on what their value of Δ_i is. In turn, the signal the lender receives is more precise for everyone.

Under our parametrization, the posterior mean of the lender that receives the new signal is:

$$\mathbb{E}[\theta | s'_i] = \mu + \underbrace{\frac{\sigma^2}{\sigma^2 + \sigma_\epsilon^2}}_{\equiv \lambda'} (s'_i - \mu)$$

where $\lambda' \geq \lambda$, reflecting that the new signal is more informative. The lender now places more weight on the signal and less on the prior.

The two-channel decomposition We can now derive the change in the posterior expectation as a function of the signal:

$$\mathbb{E}[\theta_i | s'_i] - \mathbb{E}[\theta_i | s_i] = \lambda'(s_i + \Delta_i - \mu) - \lambda(s_i - \mu) = \underbrace{\lambda' \Delta_i}_{\text{Signal Value}} + \underbrace{(\lambda' - \lambda)(s_i - \mu)}_{\text{Precision}} \quad (3)$$

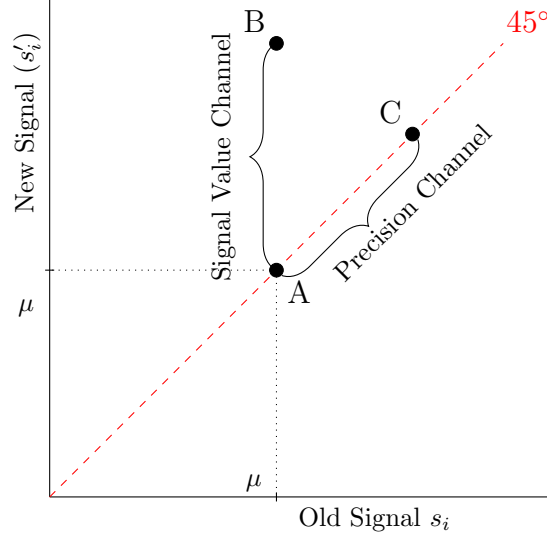
Equation 3 is central to the framework. It decomposes the policy's effect on the lender's posterior into two additive, separable terms:

1. **Signal Value Channel** ($\lambda' \Delta_i$): The revealed information Δ_i directly shifts the lender's assessment. Borrowers with positive payment histories ($\Delta_i > 0$) are upwardly revised. The magnitude scales with λ' , the precision weight of the new signal.
2. **Precision Channel** $(\lambda' - \lambda)(s_i - \mu)$: Even for borrowers whose signal *value* is unchanged ($\Delta_i = 0$), the policy raises the precision of the signal. A more precise signal pushes the lender's posterior away from the prior mean: borrowers with above-average old signals ($s_i > \mu$) are upwardly revised; those with below-average signals ($s_i < \mu$) are downwardly revised. The magnitude scales with $(\lambda' - \lambda)$, how more precise the signal became. Critically, the precision channel is zero if and only if $s_i = \mu$: borrowers whose pre-policy credit score sits exactly at the population mean experience no change in assessed creditworthiness through this channel.

We illustrate these channels in the diagram above. The x-axis represents values of the old signal (s_i), whereas the y-axis shows the new signal (s'_i) constructed with the revealed information. Points over the 45-degree line represent individuals whose signals did not change with the policy.

We highlight three points in the diagram that provide intuition for how the joint distribution maps into changes in expected creditworthiness. Point A represents an individual whose initial signal was equal to the population prior and did not change with the revealed information. Our framework predicts no change in this individual's expected creditworthiness. In turn, point B represents an individual whose signal increases with

Diagram of Joint Distribution of Old and New Signals



the revealed information. This is the case for all individuals above the 45-degree line. The opposite holds for those below the 45-degree line. Lastly, we highlight point C, which represents an individual whose signal did not change but was above the population prior. Our framework predicts that the lenders' expectation of C's creditworthiness increases with the revelation of new information as the shrinkage factor decreases.

3.3 Mapping into Empirical Comparisons

Equation 3 provides intuition on changes in the posterior expectation given a policy that increases information embedded in the signal. However, the object of interest is changes in credit, described in equation 2. Thus, we need to map the joint distribution of signals to changes in credit. In Appendix C we provide a set of assumptions necessary to do so. Here, as we are focused on simplified assumptions to provide intuition, we assume a simple linear relation in $g(E[\theta|s]) = \Gamma + \gamma E[\theta|s]$. This implies we can then write equation 2 as:

$$\Delta C_i = \alpha \cdot \Delta_i + \beta \cdot (s_i - \mu) + \varepsilon_i \quad (4)$$

Where $\alpha = \gamma\lambda'$, and $\beta = \gamma(\lambda' - \lambda)$. They are both reduced-form composites: α absorbs the product of the lender's credit response and the new precision weight, while β absorbs the lender's credit response time, the increase in the signal's precision.

The structural Equation (4) depends on two objects: $\Delta_i = s'_i - s_i$ and $s_i - \mu$, both of which are fully determined by the pair (s_i, s'_i) . Choosing a control group is therefore

equivalent to choosing a point in the joint distribution of (s_i, s'_i) . Equation (4) makes clear that the only point in this joint distribution for which we expect no change in credit through either channel is $(s_i, s'_i) = (\mu, \mu)$: the signal value term vanishes because $\Delta_i = s'_i - s_i = 0$, and the precision term vanishes because $s_i - \mu = 0$. This is the unique point in the joint distribution of (s_i, s'_i) at which the policy has no effect on the lender's posterior and, therefore, no effect on credit.

Total effect. The total effect of the policy for any point $(s_i - \mu = x, \Delta_i = d)$ in the joint distribution, relative to the (μ, μ) control group, is identified by:

$$\mathbb{E}[\Delta C_i \mid \Delta_i = d, s_i - \mu = x] - \mathbb{E}[\Delta C_i \mid \Delta_i = 0, s_i = \mu] = \alpha \cdot d + \beta \cdot x \quad (5)$$

This comparison recovers a combination of signal-value and precision effects, weighted by the average change in signal value d and the average distance from the mean x of the treated group. While this total effect is of independent interest, the framework also provides the right comparisons to identify each channel separately.

Separating the two channels: To separately identify α and β we need two comparisons, each shutting down one channel.

Signal value effect. Restrict both the treated and control groups to $s_i = \mu$, and compare $\Delta_i = d$ against $\Delta_i = 0$:

$$\mathbb{E}[\Delta C_i \mid \Delta_i = d, s_i = \mu] - \mathbb{E}[\Delta C_i \mid \Delta_i = 0, s_i = \mu] = \alpha \cdot d \quad (6)$$

With $s_i = \mu$ for both groups, the precision term $\beta(s_i - \mu)$ vanishes on both sides and the comparison recovers $\alpha \cdot d$.

Precision effect. Restrict both groups to $\Delta_i = 0$, and compare $s_i - \mu = x > 0$ against $s_i = \mu$:

$$\mathbb{E}[\Delta C_i \mid \Delta_i = 0, s_i - \mu = x] - \mathbb{E}[\Delta C_i \mid \Delta_i = 0, s_i = \mu] = \beta \cdot x \quad (7)$$

With $\Delta_i = 0$ for both groups, the signal value term $\alpha \cdot \Delta_i$ vanishes on both sides and the comparison recovers $\beta \cdot x$.

Departing from Linear Case: The linear model outlined above is useful to provide simplified objects that map into our empirical analysis. The key intuition we convey above is how to use the different parts of the joint distribution of new and old signals to recover the total effects of the more informative signal on credit access, and to separately

identify the two effects proposed in our framework. In our empirical analysis that follows, we depart from the linear case and use more flexible difference in differences models to estimate the effects of the policy. In Appendix C, we formalize the comparisons described above without the linear case, and propose more flexible ways to think through the empirical objects estimated below.

4 Effects on Credit Access

In this section, we describe the effects of Cadastro Positivo on credit access. Motivated by our conceptual framework, we explore variation caused by the policy on credit scores to identify our results. In what follows, we refer to the old and new system credit scores, which correspond to those built using the pre- and post-policy information sets, respectively (or in our framework notation (s, s')).

Our partner SERASA Experian graciously computed their standard credit scores models for our sample under their old and new systems. We show the joint distribution of credit scores in Figure 1a, where we divide the sample into 11 groups of old and new credit scores. In turn, Figure 1b shows the histogram of the difference between new and old system scores, equivalent to $\Delta_i = s'_i - s_i$ in our framework.

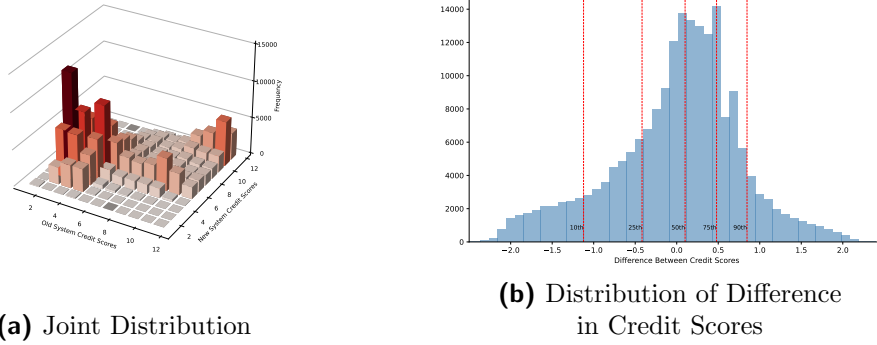
In all our analyses, we use standardized values of credit scores under the new and old systems. In Figure A4, we show the distribution of both systems' credit scores. In Figure A5, we show the correlation between credit scores and the credit and demographic characteristics of individuals. We find that women and nonwhite individuals have, on average, lower credit scores than men and white individuals. We also observe that total outstanding credit is positively correlated with credit scores in both the old and new systems. However, the correlation between both measures is substantially stronger in the new system. On average, new systems' credit scores benefited those who had a larger credit history.

4.1 Signal Value and Precision Effects

We begin by separately estimating the two channels through which the policy affects credit access. Both exercises share the same empirical structure: we define three groups in the joint distribution (positively exposed, negatively exposed, and a control group) and estimate the following difference-in-difference model:

$$Y_{it} = \alpha_i + \delta_t + \sum_{t \in T} \beta_t^+ \cdot D_i^+ \cdot \delta_t + \sum_{t \in T} \beta_t^- \cdot D_i^- \cdot \delta_t + \varepsilon_{it} \quad (8)$$

Figure 1: Joint Distribution of Credit Scores in Positive and Negative Systems



In Panel (a), we show a histogram of the joint distribution of credit scores in the new and old systems. Individuals are divided in a grid of 11 equally sized bins of values in each system. In Panel (b), we show a histogram of the difference between the z-scores of credit scores in the new system and credit scores in the old systems. Vertical red lines represent the 10th, 25th, 50th, 75th and 90th percentile of the distribution.

where α_i are individual fixed effects and δ_t are time dummies. Our identifying assumption is that, on average, individuals in different groups would follow parallel trends in the absence of the policy.

The two exercises differ only in how the groups are defined. Our goal is to mimic the comparisons outlined in the conceptual framework. To do so empirically, we approximate groups according to their position in the joint distribution of credit scores as illustrated in Figure 2.

Formally, to estimate the signal value effect, we define:

$$D_i^+ = \mathcal{I}[\Delta_i \in [0.75, 1.25]] \quad D_i^- = \mathcal{I}[\Delta_i \in [-1.25, -0.75]] \quad D_i^0 = \mathcal{I}[\Delta_i \in [-0.25, 0.25]]$$

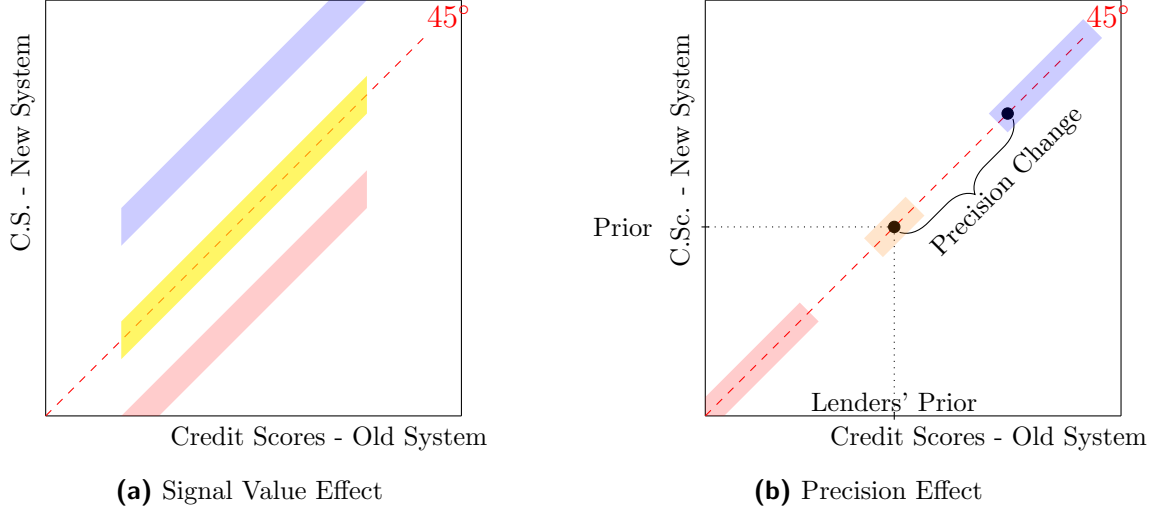
Then, we estimate Equation 8 in the restricted case of individuals lying in one of the three groups (i.e., $D_i^+ = 1 | D_i^- = 1 | D_i^0 = 1$). Note that, in this case, the sample selection satisfies the stronger assumption of matching the average pre-policy signal between treatment and control, highlighted in the previous section. This allows us to gain statistical power while still recovering unbiased estimates of the signal value.

To estimate the precision effect, we restrict the sample to individuals with $\Delta_i \in (-0.25, 0.25)$ (those whose signal value was essentially unchanged) and compare them with those far above or below the population means. In this case, the comparison groups

are formally defined as:

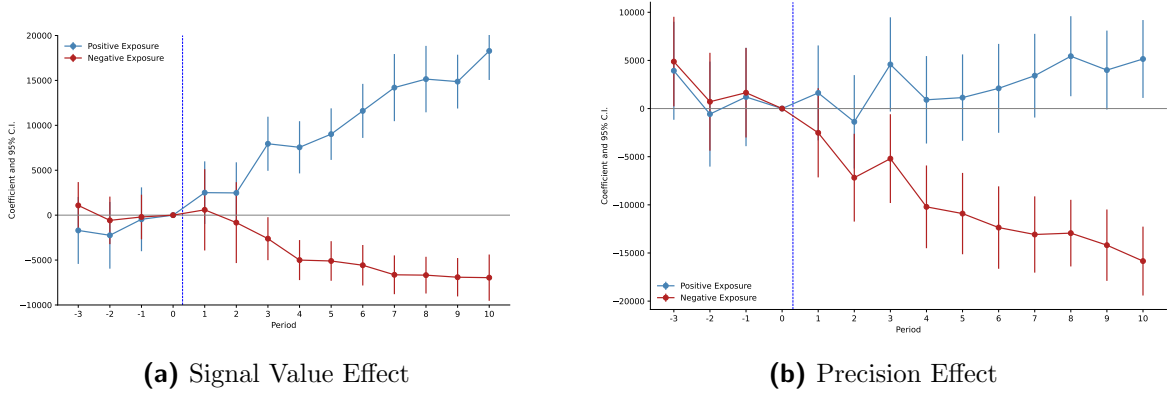
$$D_i^+ = \mathcal{I}\left[s_i > 1\right] \quad D_i^- = \mathcal{I}\left[s_i < -1\right] \quad D_i^0 = \mathcal{I}\left[s_i \in [-0.25, 0.25]\right]$$

Figure 2: Diagram of the Empirical Strategy



Results are shown in Figure 3, with Panel (3a) displaying the signal value estimates and Panel (3b) the precision estimates. The signal value effect is large and symmetric. Two years after the policy, individuals with a one-standard-deviation increase in their credit score received approximately 15,000 BRL more in credit, a 20% increase relative to their counterfactual. Those with a comparable decrease experienced a 20% reduction. The precision effect is also significant and operates as the framework predicts: individuals with above-average pre-policy scores but no change in signal value received around 5,000 BRL more in credit, while those with below-average scores lost around 12,000 BRL, despite their assessed creditworthiness remaining unchanged by the new information.

Figure 3: Difference in Differences Results



This Figure shows $\{\beta_t^+, \beta_t^-\}$ estimates from equation 8 with their respective 95% confidence intervals for both signal and precision effects exercises. In Panel (a), positively exposed individuals are those with $\Delta_i \in [0.75, 1.25]$ and negatively exposed are those with $\Delta_i \in [-1.25, -0.75]$, whereas those in the omitted control group have $\Delta_i \in [-0.25, 0.25]$. In Panel (b), the sample is restricted to individuals with $\Delta_i \in [-0.25, 0.25]$. Positive exposure are individuals with $s_i > 1$ and negative exposure corresponds to individuals with $s_i < -1$.

4.2 Overall effects and decomposition

The previous exercises isolate each channel at specific points in the joint distribution of credit scores. We now estimate how credit changes across the full distribution, without imposing functional form restrictions on how the effect varies across (s_i, s'_i) .

We divide the joint distribution into a 5×5 grid of equally sized bins. The sample is divided into five groups of equal size based on the z-score of the old system credit score, with groups defined by $k \in K = \{1, 2, 3, 4, 5\}$, where $D_i^k = 1$ if individual i belongs to group k . Similarly for the new system, with groups $j \in J = \{1, 2, 3, 4, 5\}$ and $D_i^j = 1$ if individual i belongs to group j . We estimate:

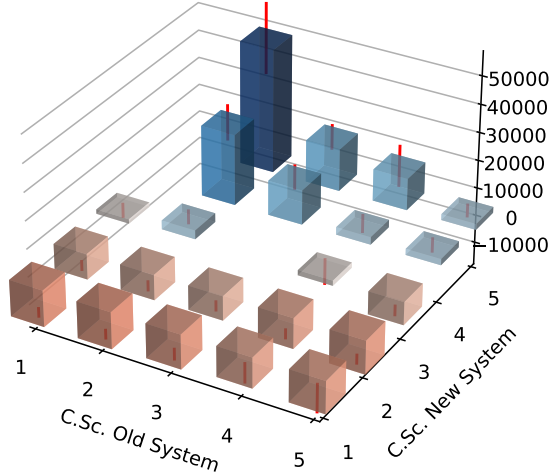
$$Y_{it} = \alpha_i + \delta_t + \sum_{k \in K} \sum_{j \in J} \gamma^{kj} \cdot D_i^j \cdot D_i^k \cdot Post_t + \varepsilon_{it} \quad (9)$$

where once more α_i are individual fixed effects, δ_t are time dummies and $Post_t$ is an indicator function that takes value one in periods after the policy ($t > 0$).

We omit $D^3 \cdot D^3$ as the reference group, the cell where the framework predicts neither channel is active, since both $\Delta_i = 0$ and $s_i = \mu$. Each γ^{kj} is therefore a difference-in-differences estimate of the credit change for individuals in that cell of the joint distribution, relative to individuals at $(s_i, s'_i) = (\mu, \mu)$. Our identifying assumption is that, absent the policy, credit in each cell would have followed the same trend as the reference group.

Figure 4 presents each estimated γ^{kj} coefficient. First, comparing estimates *vertically* (fixing the old score and moving along the new scores), credit changes increase monotonically with the new system score, reflecting the signal-value channel: a higher new score implies a larger upward revision of the lender’s posterior. Second, comparing *horizontally* (fixing the new score and moving along old scores), credit changes decrease, consistent with the signal value out-weighting the precision value for an identical absolute change in both channels. Third, comparing *diagonally* (cells with the same Δ_i but different levels of initial score), credit increases with the level of s due to the precision channel. Figure A9 plots linear fits along each of these three directions, and Table A2 reports the corresponding slope estimates.

Figure 4: Estimates of Change in Credit over the Joint Distribution of Credit Scores

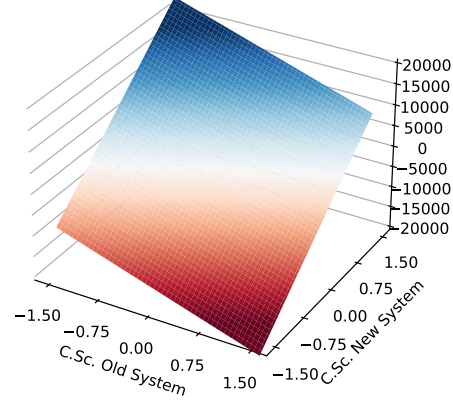


This Figure shows our estimates of changes in credit over the joint distribution of credit scores, i.e., the estimates of coefficients $\{\gamma^{kj}\}$ from equation 9. Each bar corresponds to a given coefficient, with 95% confidence intervals plotted in the red lines. Standard Errors are clustered at the individual level. Bars are organized such that the x-axis (labeled C. Sc. old system) indexes coefficients for a given group k , and the y-axis (labeled C.Sc. new system) indexes coefficients for a given group j . Positive estimates of γ^{kj} are shown in blue, whereas negative estimates are shown in red. γ^{14}, γ^{15} are not defined because there is no individual in the sample in those groups of the joint distribution of credit scores.

To summarize the average magnitude of each channel, we project the changes in credit access into the joint distribution of credit scores. In the main body of the paper, we show the results from a linear approximation, following our simplified conceptual framework. This implies estimating the following equation:

Figure 5: Estimates of Changes in Credit over the Joint Distribution with Linear $h(s_i, s'_i)$

	(1)
	Credit
C. Sc. Old Sys.	- 5,172.3 (273.66)
C. Sc. New Sys.	10,951.58 (370.70)
Observations	2875942



This Figure shows the estimates of changes in credit with linear restriction. In the Table on the left-hand side, we show our estimates for the coefficient and the standard deviation of β_0, β_1 from equation 10. On the right-hand side, the figure shows the predicted changes in credit which consists of $\hat{y}_i = \hat{\beta}_0 \cdot \text{C. Sc. Old sys.}_i \cdot \text{Post}_t + \hat{\beta}_1 \cdot \text{C. Sc. New sys.}_i$.

$$Y_{it} = \alpha_i + \delta_t + \beta_0 \cdot s_i \cdot \text{Post}_t + \beta_1 \cdot s'_i \cdot \text{Post}_t + \varepsilon_{it} \quad (10)$$

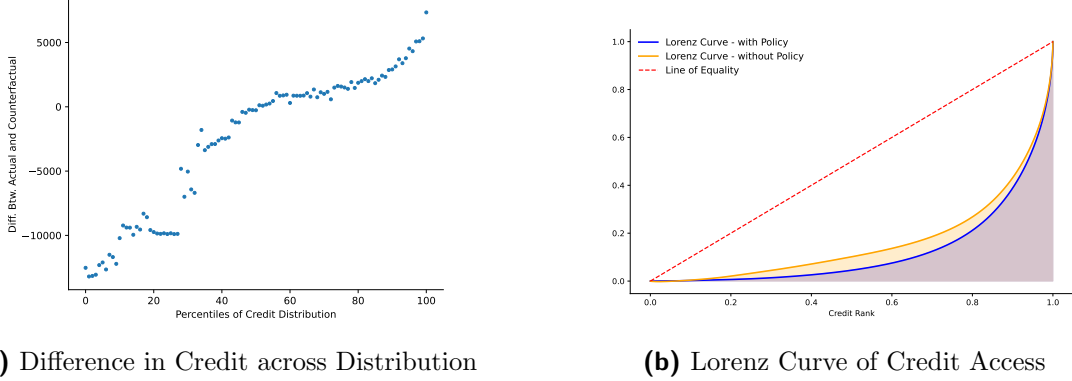
where s_i and s'_i are the z-scores of the old and new system credit scores. This is not an additional assumption relative to the conceptual framework. Notice that Equation 2 already delivers a linear relationship between ΔC_i and (s_i, s'_i) . What the semi-parametric exercise establishes is that this linear shape is a reasonable description of how the policy operated across the full distribution. In Appendix D, we provide non-parametric results using sieve-estimators.

Our findings are presented in Figure 5. The coefficient β_1 captures the signal-value channel: a one-standard-deviation increase in the new system score raises credit by 10,950 BRL, a 20% increase relative to the average individual. The precision channel is recovered from $\beta_0 + \beta_1$: a one standard deviation shift along the 45-degree line, same proportional change in both scores, so $\Delta_i = 0$, raises credit by 5,780 BRL, a 10% increase.

4.3 Distributional consequences

The policy does not affect all individuals equally, as it reallocates credit across the joint distribution of (s_i, s'_i) , and the winners and losers are not randomly distributed across the income or demographic distribution. We now characterize these distributional conse-

Figure 6: Comparisons between Distributions of Credit



This Figure shows comparisons between distributions of credit with and without the policy $\mathcal{G}_0(C_0)$ and $\mathcal{G}_1(C_1)$. In Panel (a), we plot the average difference $C_{i1} - C_{i0}$ across percentiles of credit distribution. In Panel (b), we show Lorenz curves of the actual and counterfactual credit distributions.

quences in two steps: first across the overall credit distribution, then across demographic groups.

To compare the distribution of credit with and without the policy, we construct for each individual their counterfactual credit access in the absence of the policy. Using our estimates from Equation 9, individual credit with and without the policy in the post period are defined as:

$$C_{it0} = \hat{\alpha}_i + \hat{\delta}_t \quad C_{it1} = \hat{\alpha}_i + \hat{\delta}_t + \hat{\gamma}^{kj} \cdot D_i^k \cdot D_i^j$$

where we average across post-policy periods to obtain a single value per individual.⁸ Figure 6 summarizes the comparison between the two distributions. Panel (a) shows that credit changes are almost monotonically increasing across the percentiles of the credit distribution; those with more credit before the policy gained more from it. Panel (b) confirms this through Lorenz curves: the distribution of credit without the policy lies closer to the line of equality. The variance of the credit distribution increases by 2.5% due to the policy.

We also characterize how the policy affected individuals differently according to their demographic characteristics. Our analysis is based on two dimensions in an Oaxaca-Blinder spirit. First, estimated treatment effects can be different for each group. Second, the distributions of groups can vary across the joint distribution of credit scores. Thus, even if effects are homogeneous across groups, the policy can have different aggregate

⁸For pre-policy periods $C_{i0} = C_{i1}$ since $\text{Post}_t = 0$.

effects due to compositional differences.

First, we estimate heterogeneous effects of the policy across groups. We do so by augmenting Equation 9 by interacting the cell dummies with a group indicator $G_i \in \{0, 1\}$:

$$Y_{it} = \alpha_i + \delta_t + \sum_{k \in \mathcal{K}} \sum_{j \in \mathcal{J}} \gamma_{g0}^{kj} \cdot D_i^k \cdot D_i^j \cdot \text{Post}_t + \sum_{k \in \mathcal{K}} \sum_{j \in \mathcal{J}} \gamma_{g1}^{kj} \cdot D_i^k \cdot D_i^j \cdot \text{Post}_t \cdot G_i + \varepsilon_{it}. \quad (11)$$

where γ_{g0}^{kj} captures the effect of the policy for the baseline group in cell (k, j) , and γ_{g1}^{kj} captures the differential effect for $G_i = 1$.

Using our potential outcomes notation, we can write the difference in policy effects between the two groups as:

$$\begin{aligned} \mathbb{E}[C_{i1} - C_{i0} | G_i = 1] - \mathbb{E}[C_{i1} - C_{i0} | G_i = 0] &= \frac{\sum_i G_i \cdot D_i^k \cdot D_i^j (\hat{\gamma}_{g0}^{kj} + \hat{\beta}_{g1}^{kj})}{\sum_i G_i} \\ &\quad - \frac{\sum_i (1 - G_i) \cdot D_i^k \cdot D_i^j \hat{\gamma}_{g0}^{kj}}{\sum_i (1 - G_i)} \end{aligned} \quad (12)$$

The expression above corresponds to the total effect of the policy on the gaps between groups. To isolate the composition effects, we replace group-specific coefficients with the pooled γ^{kj} from Equation 9, holding treatment effects constant at the population average.⁹

Table 2 presents our decomposition exercise results across four demographic dimensions. The results for gender are small and not economically meaningful in either direction. The most striking finding is for race: the policy increased racial inequality in credit access by approximately 4,600 BRL in favor of white individuals. This reflects both channels. Nonwhite individuals are overrepresented in the lower parts of the joint distribution of credit score (the composition effect) and, within cells, their credit gains are smaller than those of white individuals (treatment effect heterogeneity component). Both are visible in Figure A21. The pattern in education mirrors that of race: less-educated individuals are concentrated in the lower end of the distribution and receive smaller gains within cells. For age, older individuals benefit more on net, though treatment effects are more favorable to younger individuals.

These findings relate to a large body of literature that studies the effect of infor-

⁹In Appendix E we show the full derivation of our Exercise, as well as the estimates for all $\hat{\gamma}_{g0}^{kj}, \hat{\beta}_{g1}^{kj}$ coefficients, and the distribution of groups over the joint distribution of credit scores.

Table 2: Effects on Credit Differences Across Groups

	(1) Composition	(2) Total
<i>Gender</i>		
Women - Men	317.88	-745.18
<i>Race</i>		
Nonwhite - White	-2034.85	-4629.11
<i>Education</i>		
High School - Less than H.S.	1167.26	2659.22
Some College - Less than H.S.	6089.81	17462.14
<i>Age</i>		
< 40 y.o. - $\geq$ 40 y.o.	-3002.64	-1888.03

This Table shows our estimates of how the differences in credit access between groups change due to the policy. When looking at education, we add a second set of coefficients $\gamma g 2^{kj}$ and estimate equation 11 using the full sample. Column (1) shows the compositional role on the total effect by fixing treatment effects for the average population estimate (γ^{kj}) from equation 9.

mation on outcomes across races (Aigner and Cain, 1977; Arnold et al., 2022; Bartik and Nelson, 2025). On credit markets, Blattner and Nelson (2024) argue that reducing noise in credit scores would disproportionately benefit Black borrowers in the U.S. Our setting differs in an important way: the policy is theoretically race-neutral, expanding information for both white and nonwhite individuals. Yet because nonwhite individuals are disproportionately concentrated in the lower parts of the credit score distribution, a race-neutral increase in information precision mechanically widens the racial gap. This highlights a general point: the distributional consequences of information policies depend not only on what the new information says, but on where the affected populations sit in the pre-existing score distribution.

5 Effects on the Efficiency of Reallocated Credit

We documented how the policy reallocated credit across the joint distribution of credit scores and showed that this reallocation widened credit inequality and increased gaps across demographic groups. Whether these distributional consequences reflect an improvement or deterioration in credit allocation depends on the quality of the reallocation: did the policy shift credit toward borrowers who were more or less likely to default? We

now address this question by examining default rates on the credit reallocated by Cadastro Positivo.

Evaluating the efficiency effects of the policy requires characterizing two margins. First, the default rate of the additional credit extended because of the policy. Second, the default rate of the credit that would have been extended in the absence of the policy but was displaced because of it. Ideally, this exercise would rely on loan-level information linking individual loans to subsequent defaults. However, as discussed in Appendix B, our confidentiality agreement does not allow us to observe loan-level outcomes. Instead, we observe total outstanding credit and total financial delinquency amounts at the individual level.

Using these measures, we proceed in three steps. First, we estimate the effects of the policy on financial delinquency. Second, we characterize how default rates vary across the joint distribution of old and new credit scores. Third, we combine these estimates to calculate the relevant default margin of the credit reallocated because of Cadastro Positivo.

5.1 Effects on Financial Delinquency

We proceed with the same research design as in the previous section, estimating the effects of the signal’s value, the signal’s precision, and the effects across the joint distribution of credit scores (equation 9). Now, instead of total credit as the outcome, we use the total amount of financial delinquency, including zeros for individuals with no reported financial delinquency in a given period. Figure 7 shows our findings for the three different empirical strategies.¹⁰

Panel (a) shows the difference in differences estimates of the signal value effects. We observe that prior to the policy, there are no differences in the trend of financial delinquency across groups. After the policy, we find that those who were positively affected increased their total financial delinquency by almost 3000 BRL 2 years after the policy. On the other hand, individuals with negative exposure reduced financial delinquency by around 1000 BRL. Panel (b) shows the estimates of the precision effects. We see that positively exposed individuals had, on average, 1800 BRL more financial delinquency two years after the policy. We did not observe significant changes for those negatively affected. When looking at the policy’s effects on financial delinquency across the joint distribution of credit scores, we observe that most of the changes in financial delinquency are concentrated in the lower end of the distribution of old-system credit

¹⁰Figure A10 shows a two-dimensional heatmap of the estimates from Panel (c) of Figure 7.

scores.¹¹

5.2 Default Rates

Financial delinquency tends to rise among groups that experience an increase in credit access. However, as credit availability rises, the amount for which an individual can be delinquent also increases. A more policy-relevant question is what fraction of the credit extended ultimately becomes delinquent. To answer this question, we first calculate average default rates over the joint distribution of credit scores. We define default rates as the ratio between the financial delinquency of an individual at a given period and their total credit at the same period:

$$\text{Default Rate} = \frac{\text{Financial Delinquency}}{\text{Credit}}$$

Since both financial delinquency and credit change endogenously with the information revealed by the policy, we need to be careful in estimating changes in default rates. In particular, the denominator can be zero for some individuals, leading to cases where default rates are undefined.

We address this issue by using our separate estimates of financial delinquency and credit to construct default rates in the presence and absence of the policy. It is useful to define these objects using potential outcomes notation:

$$\begin{aligned} DR_{it}^0 &= \text{Default Rate}_{it}|\text{no policy} \approx \frac{E[\text{Financial Delinquency}_{it}|\text{no policy}]}{E[\text{Credit}_{it}|\text{no policy}]} \\ DR_{it}^1 &= \text{Default Rate}_{it}|\text{with policy} \approx \frac{E[\text{Financial Delinquency}_{it}|\text{with policy}]}{E[\text{Credit}_{it}|\text{with policy}]} \end{aligned}$$

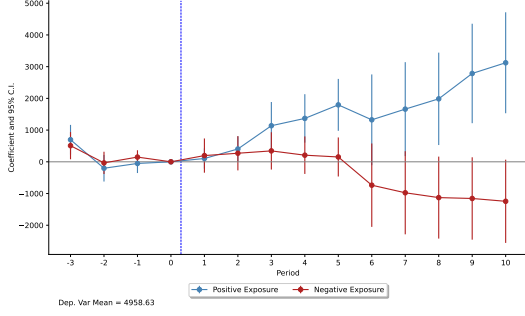
We use our estimates of the policy effects on financial delinquency and credit from our semi-parametric approach over the full joint distribution of credit scores to construct the expected values of both measures with and without the policy.¹²

For each outcome, the corresponding expectations with and without the policy are

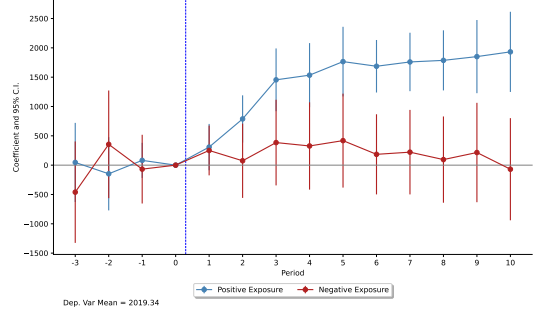
¹¹Although the patterns are less clear than in the credit access results, financial delinquency increases when comparing changes across groups *vertically*, except for those in the first group of old system credit scores, and increases through *diagonal* comparisons. When comparing estimates *horizontally*, we observe that for levels below the average new system credit score, estimates increase. Figure A11 plots linear fits between coefficients over *vertical*, *horizontal*, and *diagonal* comparisons, and Table A3 shows the estimates of linear fits between coefficients.

¹²That is, we estimate for both outcomes equation 9, which corresponds to $Y_{it} = \alpha_i + \delta_t + \sum_{k \in \mathcal{K}} \sum_{j \in \mathcal{J}} \beta_{kj} \cdot D_i^k \cdot D_i^j \cdot \text{Post}_t + \varepsilon_{it}$

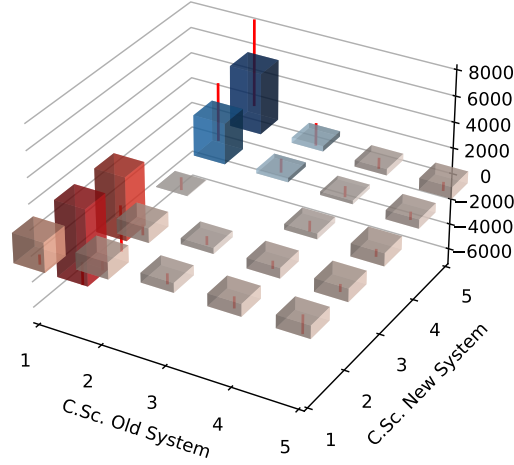
Figure 7: Effects on Financial Delinquency



(a) Signal Value Effects



(b) Precision Effects



(c) Effects over the Joint Distribution

This Figure shows estimates of the effect of the policy on Total Financial Delinquency. Individuals with no financial delinquency at a given period are included and have $Y_{it} = 0$. Panel (a) estimates the effects of the signal value, comparing individuals with $\Delta_i \in (-0.25, 0.25)$ with those positively affected $\Delta_i \in (1, 1.5)$ and negatively affected $\Delta_i \in (-1.5, -1)$. The sample is restricted to those three groups. Panel (b) estimates the effects of the signals' precision, comparing individuals with $s_i \in (-0.25, 0.25)$ with those positively affected $s_i > 1$ and negatively affected $s_i < -1$. The sample is restricted to individuals in those three groups and $\Delta_i \in (-0.25, 0.25)$. Panel (c) shows estimates of equation 9, using the full sample of individuals.

calculated as follows:

$$\begin{aligned} E[Y_{it}|\text{no policy}] &= \hat{\alpha}_i + \hat{\delta}_t \\ E[Y_{it}|\text{with policy}] &= \hat{\alpha}_i + \hat{\delta}_t + \sum_{k \in \mathcal{K}} \sum_{j \in \mathcal{J}} \gamma^{\hat{k}j} \cdot D_i^k \cdot D_i^j \cdot Post_t \end{aligned}$$

To visualize how default rates change with the policy, we aggregate them to the groups defined over the joint distribution of credit $D^k(s_i)D^j(s'_i)$. To do so, we take the average of $E[Y_{it}|\text{no policy}]$ and $E[Y_{it}|\text{with policy}]$ in each group across periods after the implementation of the policy, and calculate the ratio between expected values for financial delinquency and credit.¹³

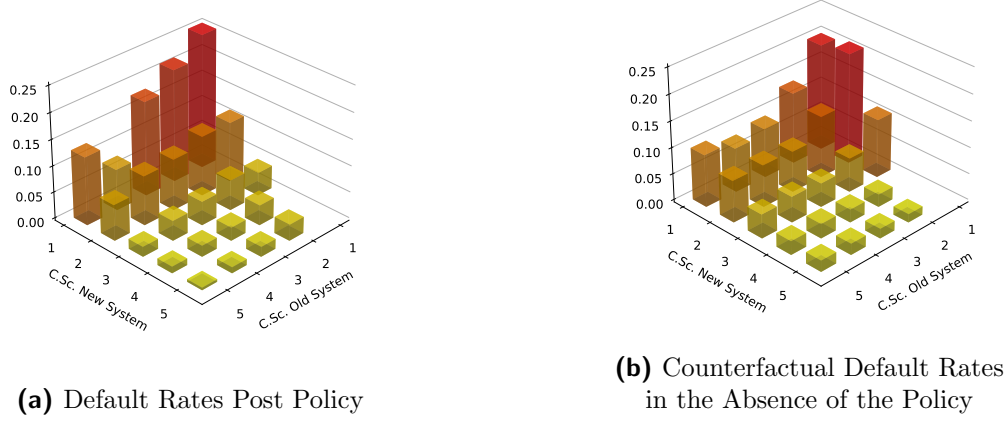
We show our findings for default rates in Figure 8. In Panel (a), we plot the default rates of each group in the presence of the policy, whereas Panel (b) shows counterfactual default rates in its absence.¹⁴ Comparing default rates with and without the policy, we see that those in the lowest levels of credit scores in the new system have an increase in their default rates, whereas those in groups 2 or higher have, on average, a decrease in their rates. Those among the higher groups of credit scores exhibit very low default rates in both systems. As seen in Figure 7, the upper parts of the joint distribution of credit scores exhibit small levels and changes in financial delinquency.¹⁵

¹³In periods before the policy $E[Y_{it}|\text{with policy}] = E[Y_{it}|\text{no policy}]$ since $Post_t = 0$

¹⁴We change the angle from the previous Figures for better visualization, but Figure A14 shows the same estimates in the previous angle.

¹⁵Differences in Default rates are shown in Figure A13.

Figure 8: Default Rates Across the Joint Distribution of Credit Scores



This Figure shows average Default Rates in the periods after the policy. Panel (a) plots our estimates of default rates in the presence of the policy, whereas Panel (b) plots our estimates of Default rates in the absence of the Policy.

5.3 Default Rates of Reallocated Credit

Having established our definition of default rates, and shown how they vary over the joint distribution of credit scores, we can estimate default rates of the reallocated credit. Under the assumption that the Cadastro Positivo did not affect default rates of credit that would have been observed in the absence of the policy, we can write the default rate of the credit allocated by the policy (DR^{Re}) as the ratio between the change in financial delinquency and changes in credit (Angrist and Imbens, 1995). To show this, it is useful to keep the potential outcomes notation. Let C_i^0, C_i^1 denote individual i 's credit absence and presence of the policy, respectively. DR_i^1, DR_i^0 are the default rates in the presence and absence of the policy, respectively.

$$DR^{Re} = \frac{\mathbb{E}[C_i^1 \cdot DR_i^1 - C_i^0 \cdot DR_i^0]}{\mathbb{E}[\Delta C_i]} = \frac{\mathbb{E}[\Delta Delinquency_i]}{\mathbb{E}[\Delta C_i]}$$

Thus, having consistent estimates of the effects of the policy on financial delinquency and credit allows us to calculate the default rate of the reallocated credit. We use our estimated coefficients from our semi-parametric analysis on delinquency and credit as approximations of the effects for each individual. We can write this as:

$$\begin{aligned}\Delta\text{Delinquency}_i &= \mathbb{E}[\Delta\text{Delinquency}_i | D_i^k = 1, D_i^j = 1] = \hat{\gamma}_{Del}^{kj} \\ \Delta C_i &= \mathbb{E}[\Delta\text{Credit}_i | D_i^k = 1, D_i^j = 1] = \hat{\gamma}_C^{kj}\end{aligned}$$

where $\hat{\gamma}_{Del}^{kj}, \hat{\gamma}_C^{kj}$ are the estimated coefficients of equation 9 using financial delinquency and credit as outcomes respectively, and D_i^k, D_i^j are dummies indicating that individual i belongs to groups k, j of the new and old system credit scores.

To compare the default rates of credit extended because of the policy with those of credit that would have been extended in its absence but is displaced by it, we divide the sample into groups with positive and negative predicted changes in credit. We then compute the average changes in financial delinquency and credit, weighting each group by its number of observations in the joint distribution of credit scores. This can be written as:

$$\begin{aligned}Positive &= \frac{\sum_i D_i^k \cdot D_i^j \cdot \hat{\beta}_{Del}^{kj}}{\sum_i D_i^k \cdot D_i^j \cdot \hat{\beta}_C^{kj}} \quad \text{for } k, j \text{ with } \mathbb{E}[\Delta\text{Credit} | D_i^k = D_i^j = 1] > 0 \\ Negative &= \frac{\sum_i D_i^k \cdot D_i^j \cdot \hat{\beta}_{Del}^{kj}}{\sum_i D_i^k \cdot D_i^j \cdot \hat{\beta}_C^{kj}} \quad \text{for } k, j \text{ with } \mathbb{E}[\Delta\text{Credit} | D_i^k = D_i^j = 1] < 0\end{aligned} \tag{13}$$

where *Positive* shows our estimates of the default rates of credit that was observed because of the policy, and *Negative* shows our estimates of the default rate of credit that would have been observed in the absence of the policy, but was not observed because of the policy.

Table 3 presents the results. On the one hand, the default rate of credit allocated because of the policy is approximately 3%. On the other hand, the default rate of credit that would have been allocated under the old system but was not allocated under the new system is approximately 15%.

Discussion: Our results indicate that the policy reallocated credit from higher-risk to lower-risk borrowers, improving the quality of the loan portfolio. At the same time, this reorientation of credit toward borrowers who are less likely to default suggests that the additional information embedded in credit scores allows lenders to better identify safer borrowers. While earlier results showed that this process is associated with greater dispersion in credit access, the evidence here suggests that the reallocation is strongly aligned with borrower risk. Taken together, the findings point to a setting in which

better information improves the composition of lending, even as it reshapes how credit is distributed across individuals.

Interpreting these effects from a policy perspective depends on how one weighs these two dimensions. From a revealed preference perspective, the reduction in credit access for higher-risk borrowers may be viewed as regressive, as these individuals are less able to borrow under the new system. At the same time, the reallocation toward lower-risk borrowers substantially reduces the default risk embedded in lending and may limit the accumulation of financial delinquency among more vulnerable individuals. While it is beyond the scope of this paper to fully evaluate these considerations, the results highlight that policies improving information in credit markets can enhance the efficiency of lending decisions, while also having meaningful distributional consequences across borrowers.

Table 3: Default Rates of Credit Reallocated with the Policy

	(1)
	Default Rate
	Reallocated Credit
Default Rate of Credit that was allocated because of the Policy	0.03
Default Rate of Credit that was not allocated because of the policy	0.15

This Table shows our estimates of the default rate of credit that was reallocated positively and negatively because of the policy. These estimates were calculated using our findings for changes in credit and financial delinquency over the joint distribution of credit scores using our expression in Equation 13.

6 Effects on Entrepreneurial Activity

A central motivation for improving credit reporting systems is that better-allocated credit should translate into more productive economic activity. The results in Section 5 show that the policy indeed improved the quality of the loan portfolio. If this reallocation were directed toward productive uses, we should expect to see effects beyond credit markets.

We focus on entrepreneurial activity, exploiting the longstanding link between credit access and the decision to start a firm (Evans and Jovanovic, 1989). Entrepreneurship is a natural margin to study for two reasons. First, starting a business is one of the most credit-intensive decisions households make, making it sensitive to changes in credit access. Second, the quality of new firms — their survival, employment, and productivity

— is a direct measure of whether credit is being channeled toward productive uses, connecting this analysis to the broader misallocation literature (Hsieh and Klenow, 2009; Restuccia and Rogerson, 2017).

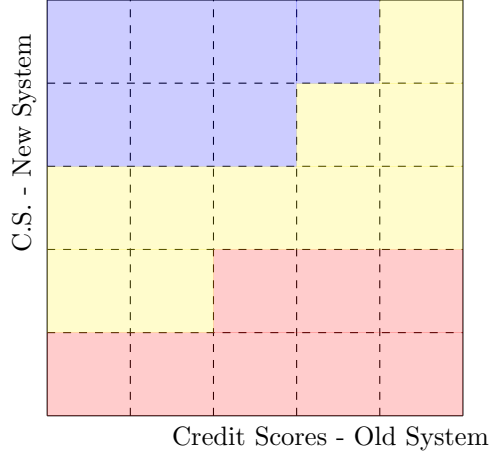
For this analysis we use the extended sample described in Section 2.2, which combines our credit bureau sample with approximately 560,000 additional individuals for whom we observe credit scores in both systems and firm ownership records from JUCESP, giving us roughly 750,000 individuals in total. This larger sample is necessary to study firm creation, which is a rare event. Table A4 confirms that the two samples are nearly identical in demographic and credit score characteristics. Details on the informality rates and job creation patterns of firms in our setting are provided in Appendices F and G.

6.1 Effects on Firm Creation

To empirically assess the policy’s effects on business creation, we slightly adjust our empirical strategy from the previous sections in two ways. First, we group individuals into three groups instead of using the 25 groups defined by the joint distribution of credit scores. We define the three groups based on our estimates of the policy’s effects on credit access. The diagram of the joint distribution of credit scores illustrates our comparison groups. Those who lie in the blue area had an increase in their credit access, while those in the red area had a decrease in their credit access. Meanwhile, individuals in the yellow area are considered our control group, as our results show that they did not have substantial changes in credit¹⁶.

¹⁶Using our definitions of the 25 groups and the same notation as before, we can define the three groups as $D^+ = \{D^{14}, D^{15}, D^{24}, D^{25}, D^{34}, D^{35}, D^{45}\}$ and $D_i^- = \{D^{11}, D^{21}, D^{22}, D^{31}, D^{32}, D^{41}, D^{42}, D^{51}, D^{52}\}$, where the first superscript refers to the group in the old system and the second the group in the new system of credit scores.

Diagram of Groups to be Compared in our Empirical Analysis



The second difference from the previous sections is that we move away from the OLS estimation of the difference in differences to a survival model, using hazard models with time-dependent covariates to estimate the probability of opening a business. We do so because individuals who establish a firm at time t are unlikely to do so again at $t + 1$. Our previous approach, using an indicator variable if the individual i opened a firm at period t as Y_{it} would incorrectly measure those at risk of creating a firm.

We focus on the probability of individual i creating their first firm at period t . We define as our *risk set*, those individuals who had never owned a firm up to 3 years before the policy. Excluding individuals who had already created firms before the 3 year window of our analysis implies our final analysis sample comprises around 650 thousand individuals. In Table A5, we summarize the demographic characteristics of our analysis sample in comparison with those who were excluded. The remaining sample is similar in age and gender composition but slightly less educated and more likely to be nonwhite.

Using our analysis sample, we construct an unbalanced panel to estimate the hazard model with time-dependent covariates. In the data, each observation corresponds to an individual x quarter. It covers six years, between the first quarter of 2018 to the last quarter of 2023. The panel is unbalanced because if an individual creates a firm, they are excluded from subsequent periods.

Our estimating equation can be written as follows:

$$\lambda(t|X(t)) = \lambda_0(t) \exp \left(\beta_0^+ \cdot D_i^+ + \beta_1^+ \cdot D_i^+ \cdot Post_t + \beta_0^- \cdot D_i^- + \beta_1^- \cdot D_i^- \cdot Post_t + \Gamma X_i \right) \quad (14)$$

where D_i^+ is an indicator that individual i is in the blue area in the diagram above, D_i^- indicates that they were in the red area, $Post_t$ is an indicator that the respective

period is after the implementation of the policy. We include a vector of controls X_i that consists of fixed effects by cells, which are defined by interacting education, gender, race and age.

The coefficients β_1^+, β_1^- capture how the probability of creating a firm changes after the policy for individuals in the positively and negatively exposed groups, relative to those in the control group who on average did not experience substantial changes in their credit access.

We show our estimates in Table 4. Both coefficients β_1^+, β_1^- are extremely close to zero and not statistically significant. Looking at hazard ratios in column (2), we show that our estimates can reject changes in the probability of opening a business as big as 5%.

Table 4: Effects on the Probability of Creating a Firm

	(1)	(2)
	Created a Firm	
	Coefficient	Hazard Ratio
Positive Exposure x Post	0.004104 (0.0169)	1.004 [0.971,1.038]
Negative Exposure x Post	-0.0179 (0.0119)	0.982 [0.959,1.006]
Number of Individuals	645040	
Observations	13170363	

This table shows coefficients and hazard ratio estimates from equation 14. In column (1), we show coefficient estimates and standard errors in parenthesis; in column (2), we show the hazard ratio and 95% confidence interval in brackets. The sample comprises all individuals we observe who were never a firm owner until three years before the policy implementation. A row in the data corresponds to sub-spells of a quarter for each individual. If the individual creates a firm, they are not observed in subsequent periods.

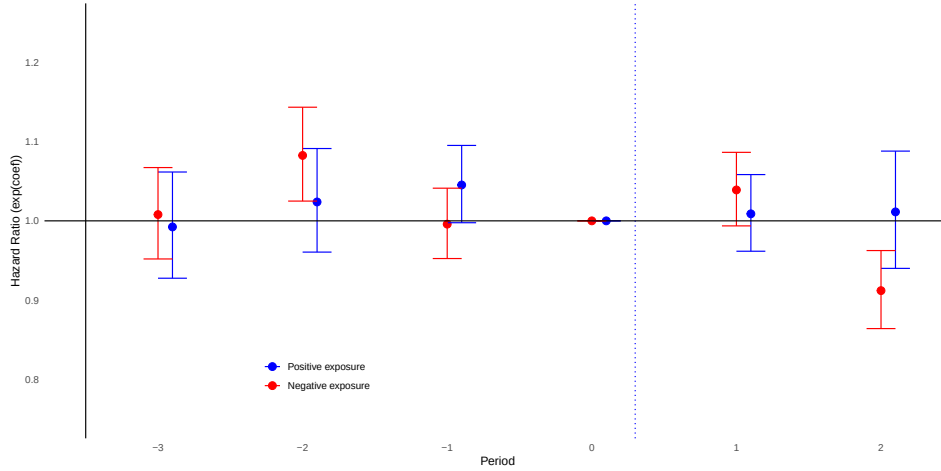
In addition to the pre-post analysis, we can also estimate coefficients for each year relative to the policy. This can be summarized in equation 15 below:

$$\lambda(t|X(t)) = \lambda_0(t) \exp \left(\beta_0^+ \cdot D_i^+ + \beta_2 \cdot D_i^- + \sum_{year \in \{-3,2\}} \beta_{year}^+ \cdot D_i^+ \cdot \delta_{year(t)} + \sum_{year \in \{-3,2\}} \beta_{year}^- \cdot D_i^- \cdot \delta_{year(t)} + \Gamma X_i \right) \quad (15)$$

where $\delta_{year(t)}$ is an indicator function that the corresponding sub-spell indexed by t is in the respective year relative to the policy implementation. We then recover coefficients $\beta_{year}^+, \beta_{year}^-$ for each year before and after the policy in our analysis.

We show our estimates of β_t^+, β_t^- in Figure 9. For the positively exposed group, our estimates indicate a precise zero in the change in the probability of opening a business. At the same time, for those negatively exposed, we find that estimates vary more across years. Despite having a negative effect 2 years after the policy (in line with what a credit constraints story would suggest), we show that one year after the policy, estimates actually suggest a positive effect of the policy for negatively exposed individuals. Pooling those estimates gets us the zero effects described above.

Figure 9: Effects on the Probability of Creating a Firm by Year



This Figure shows how the policy affected firm creation. We show the estimates of coefficients $\{\beta_t^+, \beta_t^-\}$ from equation 15. The sample comprises all individuals we observe who were never a firm owner until three years before the policy implementation. A row in the data corresponds to sub-spells of a quarter for each individual. If the individual creates a firm, they are not observed in subsequent periods.

6.2 Effects on Firm's Outcomes

We next explore if the policy affected the outcomes of firms created. Since we find no effects on firm creation, we return to the difference in differences analysis using linear

models. We restrict our sample only to eventual entrepreneurs. We then compare the outcomes of firms created by positively and negatively exposed individuals, before and after the policy, with those created by individuals in our control who had no changes in credit access.

Using our sample of entrepreneurs, we estimate the following equation:

$$Y_i = \delta_{t(i)} + \beta^+ \cdot D_i^+ \cdot Post_{t(i)} + \beta^- \cdot D_i^- \cdot Post_{t(i)} + \Gamma X_i + \varepsilon_i \quad (16)$$

where $\delta_{t(i)}$ are fixed effects of the quarter in which individual i created their firm. D_i^+, D_i^- are indicator variables that take value one if individual i belongs to groups positively or negatively affected. $Post_{t(i)}$ is an indicator variable that takes value one if the firm was created after the policy. We include a set of control variables X_i , which includes gender, race, age, education, and fixed effects of the group of the joint distribution of credit individual i belong to.

Therefore, our coefficients of interest $\{\beta^+, \beta^-\}$ capture the difference in outcomes of firms created by positively (negatively) exposed individuals before and after the policy, under the assumption that in the absence of the policy, their difference would behave similarly to the difference for control individuals.

Firm Survival: We look at firm survival one and two years after firm creation. We define a firm surviving as having an active registry in *Receita Federal*, the Brazilian equivalent of the IRS. Using public data, we can observe firm survival until August 2024 for all formal firms.¹⁷ To make it a consistent outcome, we restrict our sample to firms created until August 2022, around six quarters after the policy implementation.

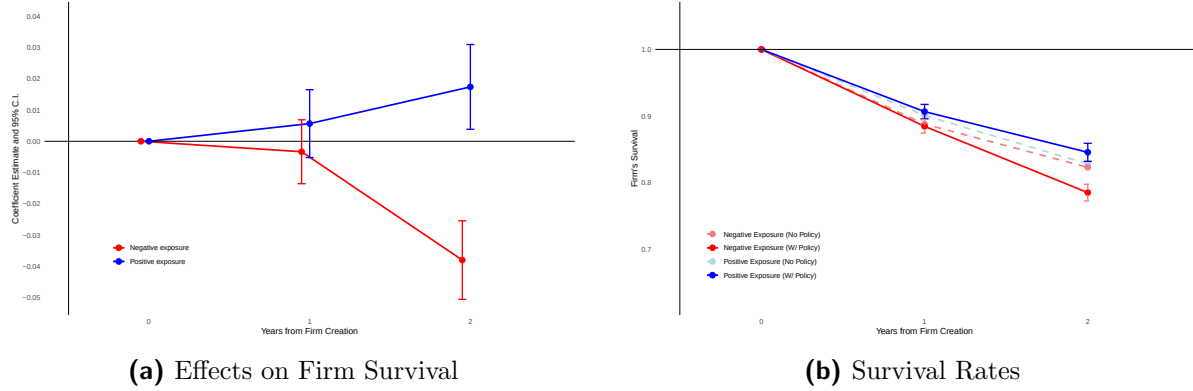
We find that more (less) credit access increases (decreases) the likelihood of firm survival two years after their creation. We show these results in Figure 10. In Panel (a) we plot our estimates and 95% confidence intervals of the coefficients β^+, β^- from equation 16. We estimate two different regressions using as outcome firm survival one and two years after firm creation. Our estimates indicate that firms created by positively exposed individuals increase by 1.8 p.p. their likelihood of survival two years after their creation because of the policy. In contrast, those created by negatively exposed individuals have a 3.8p.p. smaller likelihood of survival because of the policy.

This represents a 2.1% increase in the likelihood of firm survival and a 4.6% decrease for firms created by positively and negatively exposed individuals, respectively. We can

¹⁷We use [CNPJ](#) public data to construct firm survival information. This data is a snapshot of all active and inactive registered firms in the country, identified by their registry number (often referred to as CNPJ number). A firm is defined as inactive if it has *situação cadastral* different from *Ativa* (active). This includes firms defined as *nula*, *suspensa*, *inapta* or *baixada*. In the data, we also see the date the firm became inactive, which allows us to construct our outcome variables.

see this in Panel (b) of Figure 10. We plot the average survival of firms created by positively and negatively exposed individuals with and without the policy. Survival without the policy is calculated as the average likelihood of firms created before the policy by the respective group. Survival with the policy is defined as that value plus the coefficient estimate from Panel (a). The percent increase is calculated as the ratio between the firm survival with the policy by the firm survival without the policy.

Figure 10: Effects on Firm Survival



This Figure presents our estimates of the effects of the policy on the survival of firms created by individuals exposed to changes in credit access. The sample comprises all individuals who created their first firm three years before and two years after the implementation of the policy. In Panel (a), we show the estimates and 95% confidence intervals of coefficients $\{\beta^+, \beta^-\}$ from equation 16. In Panel (b), we show firm survival rates with and without the policy for each group. Firm survival rates without the policy are calculated as the average survival of firms created by individuals in each group before the policy. Coefficients for one-year survival and two years survival are estimated in separate regressions. Controls included are cells of education, gender, race, and year of birth, and dummies for the group of the joint distribution of credit scores the individual belonged to.

6.3 Consequences on Quality of Entrepreneurial Activity

Next, we discuss the consequences of the policy on the aggregate quality of entrepreneurial activity. We observe that changes in personal credit access cause firms' probability of survival to change. If firms that are surviving more because of the policy are of better *quality* than the ones that are not surviving because of it, then the policy increases the average quality of firms in the new cohorts of entrepreneurs.

In what follows, we compare *quality* measures of firms created by individuals across the groups of positive and negative exposure to the policy. A challenge we face is the availability of detailed firm-level data on their outcomes. Unfortunately, we do not observe revenues or other balance sheet outcomes in our firm records. We overcome this challenge in two ways. First, we use the characteristics of these firms' industries

to characterize them. This analysis is useful because we can characterize virtually all firms in our sample because we observe their detailed industry classification in our firm records.

Industry Level Characteristics: Firms created by positively exposed individuals are in better industries than the ones created by those in the control group. The opposite happens for firms created by entrepreneurs in the negatively exposed group. To show this, we calculate industry characteristics at the 5-digit CNAE level.¹⁸ We then estimate linear models of the following type:

$$Y_{ind(i)} = \delta_{t(i)} + \beta^+ \cdot D_i^+ + \beta^- \cdot D_i^- + \Gamma X_i + \varepsilon_i \quad (17)$$

where $Y_{ind(i)}$ represent characteristics of industry ind in which i 's firm is operating. In the vector X_i , we include controls for the observable characteristics of these individuals, namely fixed effects for cells of education, gender, race, and year of birth. We also add fixed effects of the period of firm creation $\delta_{t(i)}$.

Exposure to the policy is positively correlated with the employment characteristics of firms' industries. We show this in Figure 11, where we plot the coefficients β^+, β^- from equation 17 normalized by the average value of the control group.¹⁹ Firms created by individuals in the positively exposed groups are in industries that, on average, employ more and pay higher wages. This is true not only for the average firms in these industries but also for new firms.

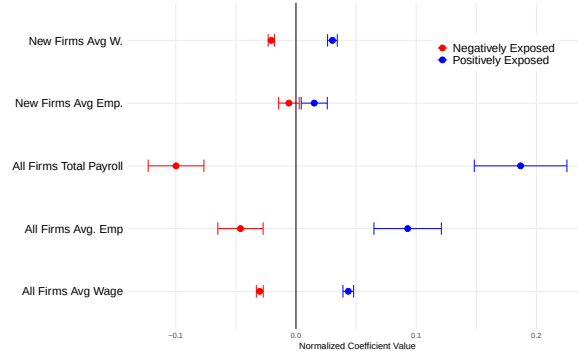
We also show that exposure positively correlates to the industries' productivity characteristics. We calculate value added per firm at the industry level²⁰ and firm fixed effects from two-way fixed effects wage regressions.²¹ We observe in Table 5 that exposure to the policy correlates positively with the productivity measures. Firms created by positively exposed individuals are 1 p.p. more likely to be in industries above the median in average value added. In contrast, those created by negatively exposed individuals are 2.5 p.p. less likely to be operating in such industries than the control group.

¹⁸CNAE is the Brazilian equivalent of NAICS industry definition. In the 5-digit level, it consists of around 700 different industries. All formal firms must report their CNAE classification.

¹⁹Table A6 shows coefficient estimates not normalized.

²⁰Value Added measures are calculated using the Brazilian Institute of Statistics (IBGE) surveys of the services, manufacturing, and construction (PIA, PAS, PAC) in 2019. These are aggregate measures in around 100 industries categories. They are less granular than the 5-digit CNAE-Level and therefore, have less variation.

²¹These consists of estimating the Abowd et al. (1999) (AKM) model where for worker i in firm j we estimate $y_{it} = \alpha_i + \phi_{j(i)} + \Gamma X_{it} + \varepsilon_{it}$. We decompose firm fixed effects for men and women as in Card et al. (2016) and aggregate them at the industry level. Bender et al. (2018) shows that firm fixed effects from wage regressions with individual and firm fixed effects are positively correlated with firm-level productivity.

Figure 11: Industry Characteristics

This Figure shows characteristics of the industries in which firms created by individuals in different treatment groups operate. They show point estimates and 95% confidence intervals of coefficients β^+ , β^- from equation 17 normalized by the average outcome value of the control group. New firm characteristics are constructed using matched employer-employee data three years after creation. "All firms" characteristics are constructed using the average value of outcomes for all firms in the given 5-digit CNAE industry classification. The sample comprises all individuals who created their first firm three years before and two years after the implementation of the policy.

Table 5: Industry Value Added and Avg. Firm F.E.

	(1)	(2)	(3)	(4)
	Value Added		Firm Wage F.E.	
	Avg.	I(Above Median)	Men	Women
Positive Exposure	2872.113 (5605.677)	0.010* (0.005)	0.009*** (0.001)	0.005*** (0.001)
Negative Exposure	-3053.021 (4477.762)	-0.026*** (0.004)	-0.004*** (0.000)	-0.001** (0.000)
Control Avg.	145811.131	0.464	-0.039	-0.054
N. of observations	83860	83860	90532	90532

This Table shows our estimates of β^+ and β^- in equation 17. Sample in columns (1) and (2) includes all firms in industries for which we were able to recover Value Added measures from the Brazilian Institute of Statistics. Value Added is calculated as the average value added of firms in the given industry. Firm wage F.E. are the average of firm-level fixed effects calculated using Card et al. (2016) model with matched employer-employee data of the universe of the Brazilian Labor Market and then aggregated at the 5-digit CNAE level.

Discussion: Our findings indicate that firms created by positively exposed groups are, on average, of higher quality than those created by individuals negatively exposed to the policy. Extrapolating the average difference to marginal firms whose likelihood of surviving changes due to the policy, our results indicate that the average quality of new cohorts of firms increases due to the policy. That is because firms that close due to the policy are of lower quality than those that survive because of it.

However, a rough back-of-the-envelope calculation suggests that overall effects on the average quality of firms are relatively small. We calculate approximations of the difference between actual and counterfactual average value added and the average number of employees. We do so by multiplying the average difference in the industry characteristics by the differential effects on firm survival. This exercise suggests that the policy increased by around 0.5% the average value added, and by 1% the average number of employees in the new cohort of entrepreneurs.

7 Conclusion

Credit reporting systems are widely advocated as tools to reduce financial frictions by addressing information asymmetries between borrowers and lenders. However, their development remains limited in many countries, with substantial variation in both their coverage and the depth of information they provide. In developing countries in particular, these systems are often underdeveloped, a shortcoming that may hinder economic development by reducing the efficiency of credit allocation.

In this paper, we present evidence on the effects of expanding the information available to lenders through an improvement in credit reporting systems. We do so by exploring a policy implemented in Brazil that transformed a system primarily based on negative financial information into one incorporating borrowers' complete credit histories into credit assessment.

Our empirical analysis shows that the effects of expanding information about borrowers can be rationalized by a simple conceptual framework that incorporates changes in both the value of the signal and its precision. We highlight this by estimating how credit access changes over the joint distribution of credit scores constructed with and without the new information made available by the policy.

We show that the reform increased efficiency in credit markets by reallocating credit from riskier to safer borrowers. At the same time, however, the policy increased overall inequality in credit access and widened racial disparities in credit between white and nonwhite individuals.

We then show that changes in credit had effects beyond credit markets by investigating the effects of the policy on entrepreneurial activity. While we do not find evidence that the policy increased the likelihood of firm creation, we show that it affected the composition and survival of firms created by cohorts exposed to the reform. In particular, firms associated with individuals who experienced increases in credit access exhibit higher survival rates and are positively correlated with characteristics typically associated with higher-quality firms, while the opposite pattern emerges for individuals whose credit access declined after the policy. These findings suggest that the reallocation of credit improved the selection of entrepreneurial projects, increasing the survival of firms associated with borrowers who became more creditworthy under the new information environment.

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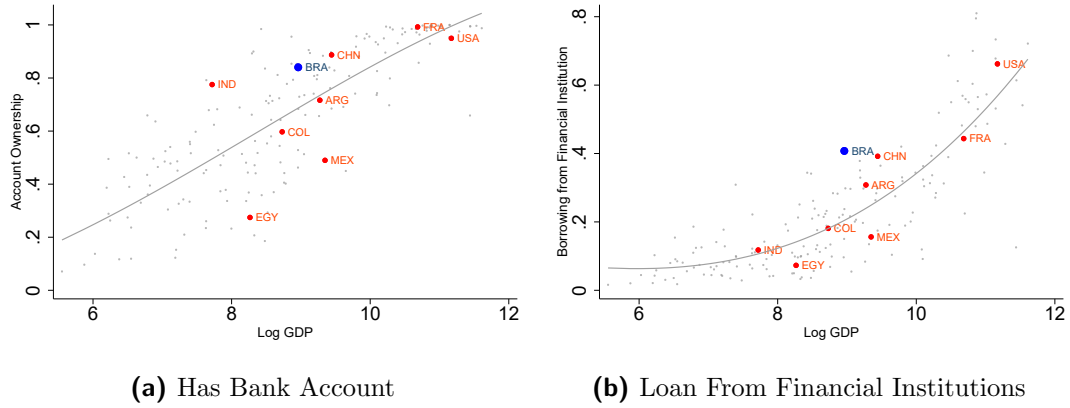
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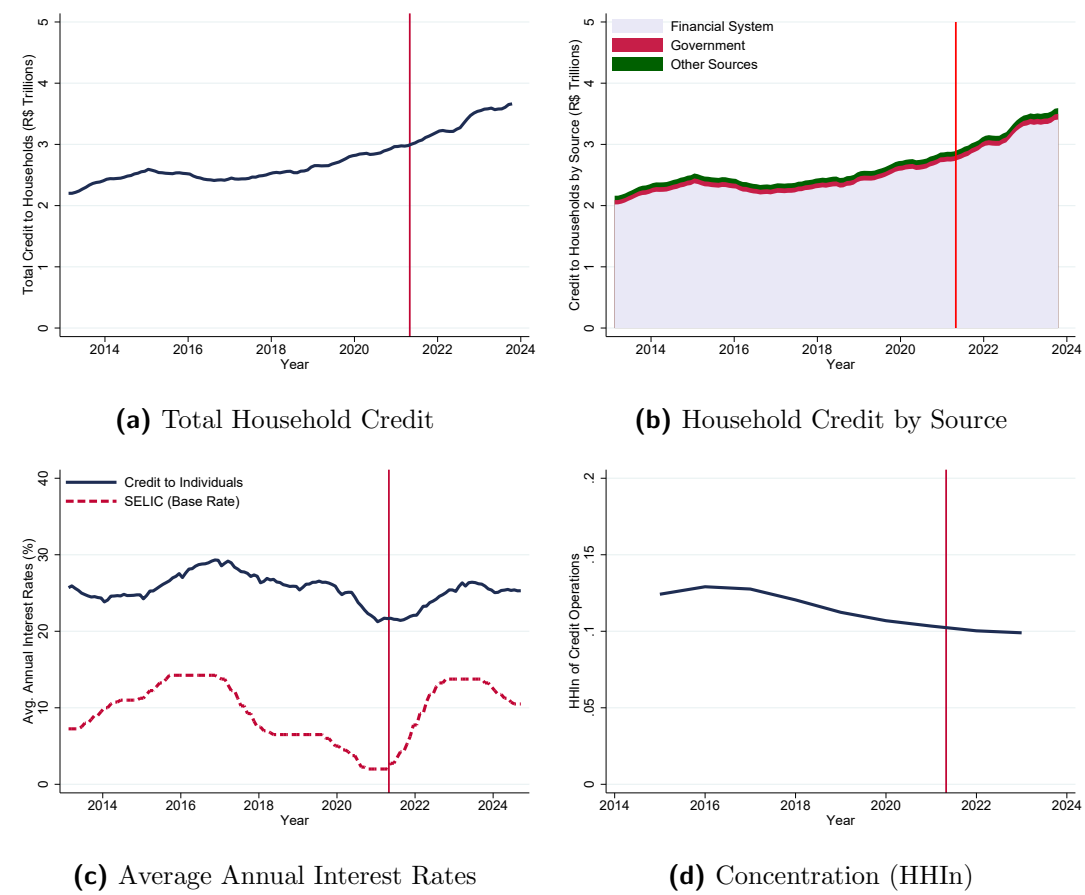
A Additional Figures and Tables

Figure A1: Credit Characteristics by Country and Income



This Figure shows the credit characteristics of countries relative to the Log of their Per capita GDP. Figures were constructed using 2021 data from the World Bank Global Financial Development Database.

Figure A2: Aggregate Credit Statistics Before and After Cadastro Positivo



This Figure shows aggregate credit statistics in the periods before and after *Cadastro Positivo* implementation, indicated by the red vertical lines. In Panel (a), we show Total Household Credit in the country. Panel (b) shows the same measure decomposed by the source of that given credit. Other Sources included credit cooperatives and institutions non directly tied to the financial system. Panel (c) plots the average interest rate of credit given to households. Panel (d) plots normalized Herfindahl-Hirschman Index. The index is obtained by summing the square of the market share (in decimal form) of each financial institution in the considered market: $HHIn = (IF1)^2 + (IF2)^2 + \dots + (IFj)^2$, resulting in a number between 0 and 1. Normalization choice was made by the Brazilian Central Bank. Panels (a), (b), and (c) use data available in the [Brazilian Central Bank's Time Series Management System](#). Panel (d) uses data collected from the [Brazilian Central Bank's report of banking activity](#).

Figure A3: Example of Firm Record from Sao Paulo's Trade Board



GOVERNO DO ESTADO DE SÃO PAULO
SECRETARIA DE DESENVOLVIMENTO ECONÔMICO
JUNTA COMERCIAL DO ESTADO DE SÃO PAULO

FICHA CADASTRAL COMPLETA



NESTA FICHA CADASTRAL COMPLETA, AS INFORMAÇÕES DOS QUADROS "EMPRESA", "CAPITAL", "ENDEREÇO", "OBJETO SOCIAL" E "TITULAR/SÓCIOS/DIRETORIA" REFEREM-SE À SITUAÇÃO DA EMPRESA NO MOMENTO DE SUA CONSTITUIÇÃO OU AO SEU PRIMEIRO REGISTRO CADASTRADO NO SISTEMA INFORMATIZADO.

A AUTENTICIDADE DESTA FICHA CADASTRAL COMPLETA PODERÁ SER CONSULTADA NO SITE WWW.JUCESPONLINE.SP.GOV.BR, MEDIANTE O CÓDIGO DE AUTENTICIDADE INFORMADO AO FINAL DESTES DOCUMENTOS.

PARA EMPRESAS CONSTITUÍDAS ANTES DE 1.992, OS ARQUIVAMENTOS ANTERIORES A ESTA DATA DEVEM SER CONSULTADOS NA FICHA DE BREVE RELATO (FBR).

EMPRESA			
Firm name			
NIRE MATRIZ		DATA DA CONSTITUIÇÃO	TIPO: SOCIEDADE LIMITADA
		15/03/2007	11/10/2021 08:52:43
INÍCIO DE ATIVIDADE	CNPJ	INSCRIÇÃO ESTADUAL	
14/03/2007			

CAPITAL	
R\$ 66.000,00 (SESSENTA E SEIS MIL REAIS)	
Initial capital	

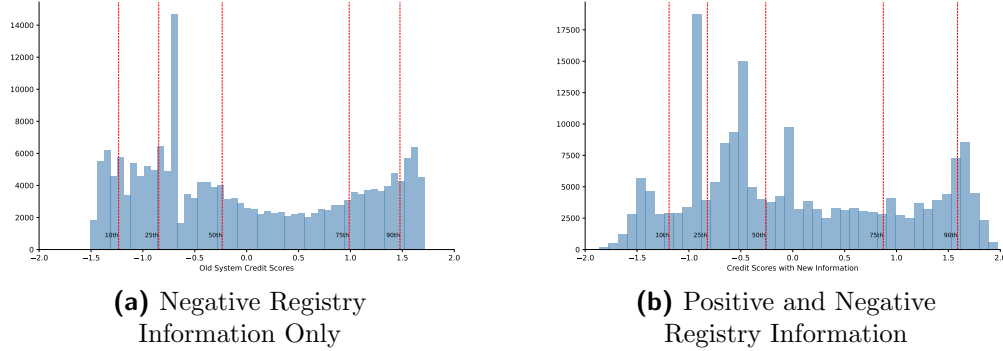
ENDEREÇO			
LOGRADOURO: ESTRADA MUNICIPAL	Street	NÚMERO:	Number
BAIRRO: BAIRRO DOS PIRES	Neighborhood	COMPLEMENTO:	
MUNICÍPIO: LIMEIRA	Municipality	CEP: 13480-000	UF: SP
		Zip Code	

OBJETO SOCIAL	
Firm industry	
INCORPORAÇÃO DE EMPREENDIMENTOS IMOBILIÁRIOS OBRAS DE TERRAPLENAGEM COMÉRCIO VAREJISTA DE MATERIAIS DE CONSTRUÇÃO EM GERAL	

TITULAR / SÓCIOS / DIRETORIA	
Full name	Nationality
IDENTIFICADOR: , NACIONALIDADE BRASILEIRA, CPF: , RG/RNE: , RESIDENTE À RUA CAPITÃO MANOEL FERRAZ DE CAMARGO, JD. PIRATININGA, LIMEIRA - SP, CEP 13484-333, NA SITUAÇÃO DE SÓCIO COM VALOR DE PARTICIPAÇÃO NA SOCIEDADE DE \$ 6.000,00	
Owner's capital	Full address
IDENTIFICADOR: , NACIONALIDADE BRASILEIRA, CPF: , RG/RNE: , RESIDENTE À RUA CAPITÃO MANOEL FERRAZ DE CAMARGO, JD. PIRATININGA, LIMEIRA - SP, CEP 13484-333, NA SITUAÇÃO DE SÓCIO E ADMINISTRADOR, ASSINANDO PELA EMPRESA, COM VALOR DE PARTICIPAÇÃO NA SOCIEDADE DE \$ 15.000,00	
	Role in the firm
IDENTIFICADOR: , NACIONALIDADE BRASILEIRA, CPF: , RG/RNE: , RESIDENTE À RUA PROFA. OTILIA	

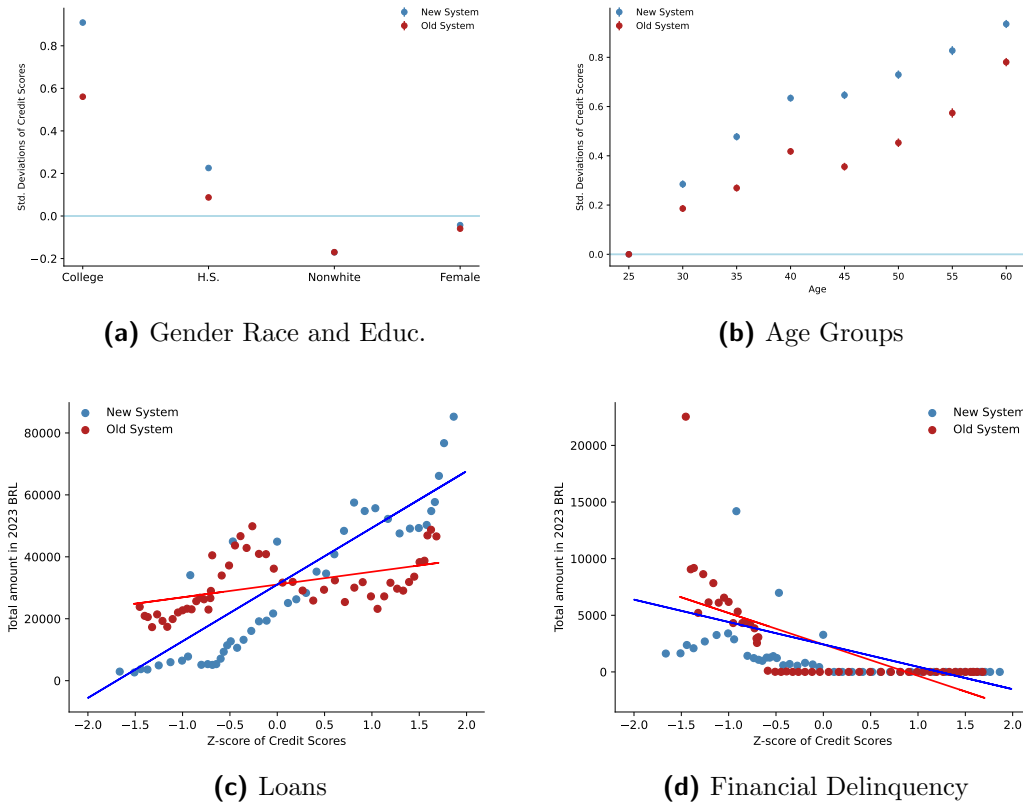
This Figure shows an example of a firm record from São Paulo's trade board. Despite being publicly available records, we blurred identifiable information of individuals and firms to protect their confidentiality. All records are available in *Serviços Online* under the option *Pesquisar Empresas*. It is necessary to have a Brazilian social security number (CPF) to log in the website. Additional information such as ownership and capital change are also available in these records despite not being apparent in this example.

Figure A4: Histogram of Credit Scores under both Systems



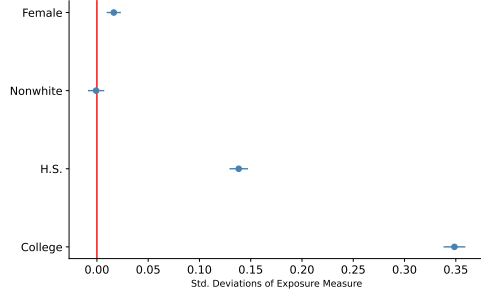
This Figure shows the histogram of credit scores calculated with information from both positive and negative registries, and credit scores calculated with information from the negative registry. The sample is restricted to the last period before the implementation of the policy.

Figure A5: Credit Scores Correlation with Observable Characteristics

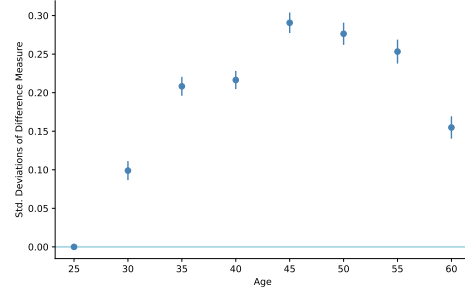


Panels (a) and (b) plot the coefficients of a regression of $\frac{cs_i - \bar{cs}}{sd(cs_i)}$ on observable characteristics. The sample is restricted to the last period before the implementation of the policy. Coefficients in panels (a) and (b) are estimated in the same regression that includes dummies for gender, race, education groups, and age groups. We omit from the regression white men with less than high school education in the youngest age group. In panels (c) and (d), we show binscatters of Credit and Default with the Z-scores of credit scores in both positive and negative systems. The sample is restricted to the last period before the implementation of the policy.

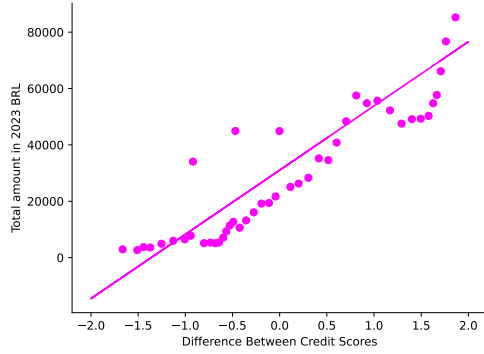
Figure A6: Correlation of the Difference between Credit Scores with Observable Characteristics



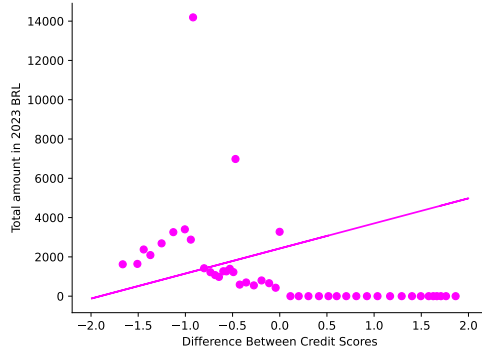
(a) Gender Race and Educ.



(b) Age Groups



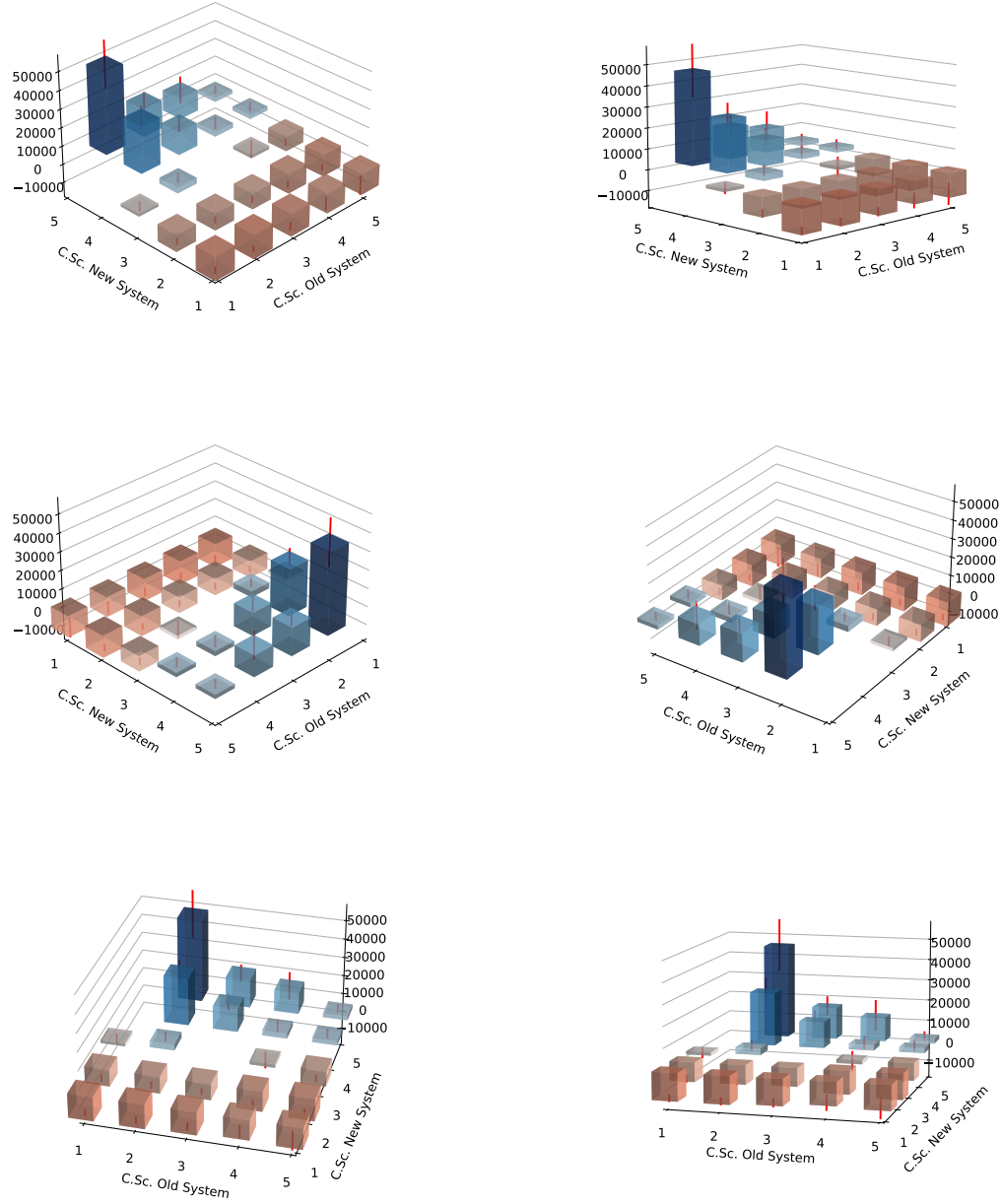
(c) Total Credit



(d) Payments not made

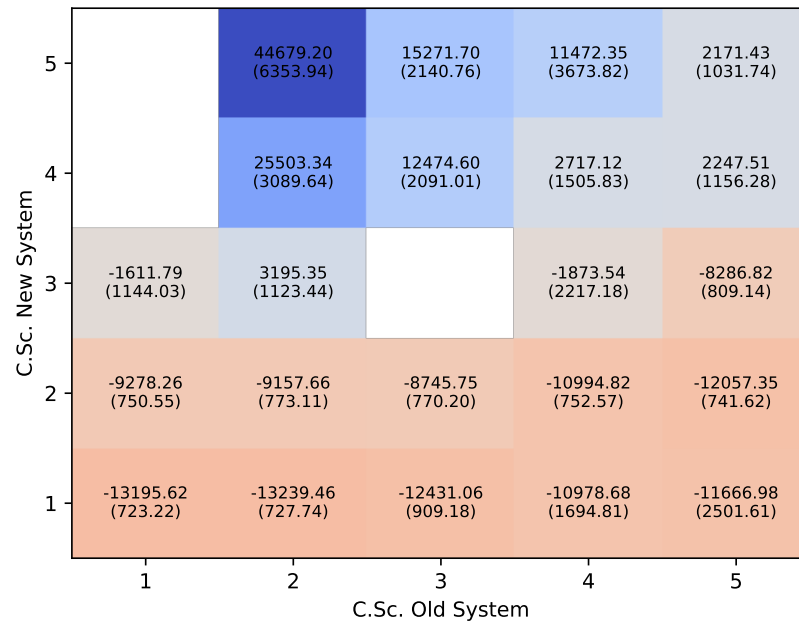
Panels (a) and (b) plot the coefficients of a regression of Δ_i on observable characteristics. The sample is restricted to the last period before the implementation of the policy. Coefficients in panels (a) and (b) are estimated in the same regression that includes dummies for gender, race, education groups, and age groups. We omit from the regression white men with less than high school education in the youngest age group. In panels (c) and (d), we show binscatters of Credit and Default with respect to Δ_i . The sample is restricted to the last period before the implementation of the policy.

Figure A7: Effects on Credit over the Joint Distribution of Credit Scores - Rotated Bars



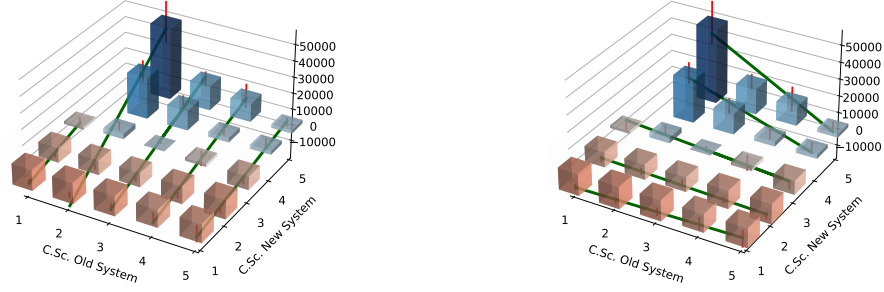
This Figure shows the estimates of coefficients $\{\gamma^{kj}\}$ from equation 9. It shows the same results as in Figure 4 but over different angles. Each bar corresponds to a given coefficient, with 95% confidence intervals plotted in the red lines. Standard Errors are clustered at the individual level. Bars are organized such that the x-axis (labeled C. Sc. old system) indexes coefficients for a given group k , and the y-axis (labeled C.Sc. new system) indexes coefficients for a given group j . Positive estimates of γ^{kj} are shown in blue, whereas negative estimates are shown in red. γ^{14}, γ^{15} are not defined because there is no individual in the sample in those groups of the joint distribution of credit scores.

Figure A8: Effects of the Policy on Credit over the Joint Distribution of Credit Scores



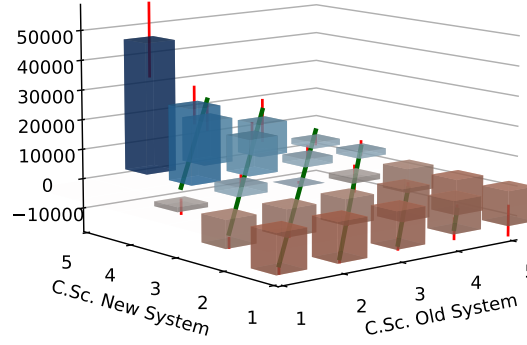
Notes: This Figure shows estimates of coefficients $\{\gamma^{kj}\}$ from equation 9. Standard errors in parenthesis are clustered at the individual level. Coefficients are organized such that the x-axis (labeled C. Sc. old system) indexes coefficients for a given group k, and the y-axis (labeled C.Sc. new system) indexes coefficients for a given group j.

Figure A9: Comparisons Between Changes in Credit over the Joint Distribution of Credit Scores



(a) *Vertical Comparisons*

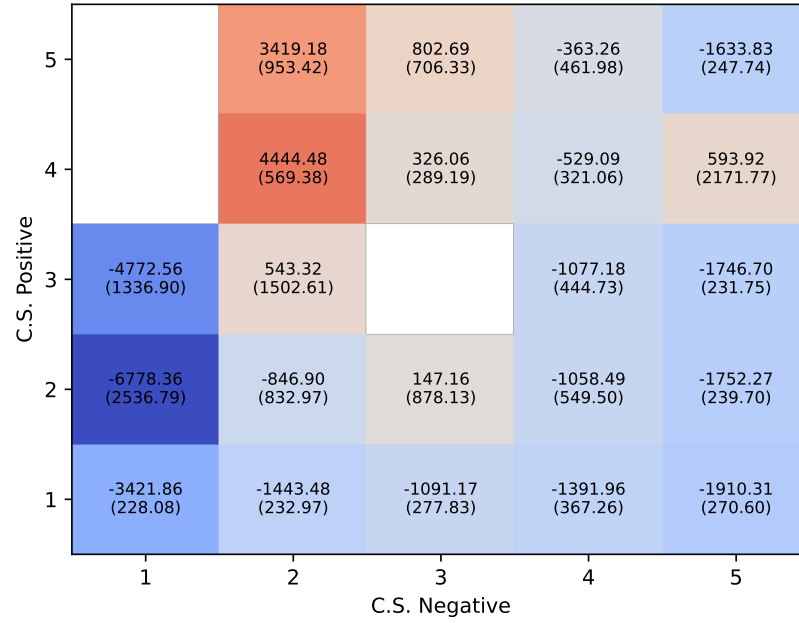
(b) *Horizontal Comparisons*



(c) *Diagonal Comparisons*

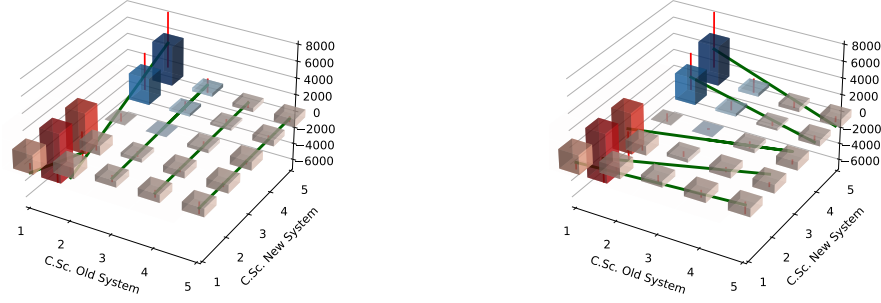
This Figure shows the estimates of coefficients $\{\gamma^{kj}\}$ from equation 9 with the same specifications as Figure 4. The difference is the green lines that correspond to the linear fit between coefficients. A linear fit is computed by regressing values of estimates along each coordinate of the corresponding comparison without weighting. Panel (c) shows the same estimates but rotated to a different angle to better visualize the diagonal comparisons. Table A2 shows the estimates of the linear fits across coordinates.

Figure A10: Effects of the Policy on Financial Delinquency over the Joint Distribution of Credit Scores



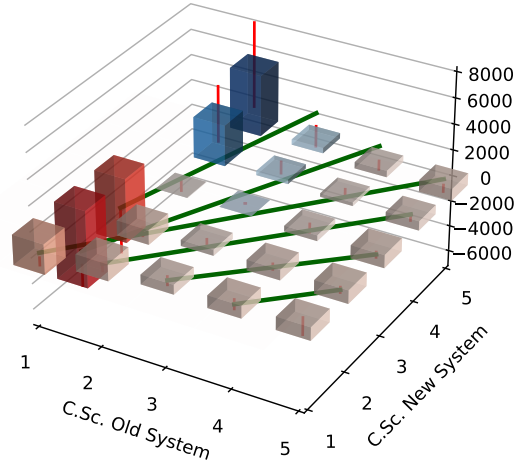
Notes: This Figure shows estimates of coefficients $\{\beta^{kj}\}$ from equation 9 with total financial delinquency from individual i at period t as the outcome. Standard errors in parenthesis are clustered at the individual level. Coefficients are organized such that the x-axis (labeled C. Sc. old system) indexes coefficients for a given group k , and the y-axis (labeled C.Sc. new system) indexes coefficients for a given group j .

Figure A11: Comparisons Between Changes in Financial Delinquency over the Joint Distribution of Credit Scores



(a) *Vertical Comparisons*

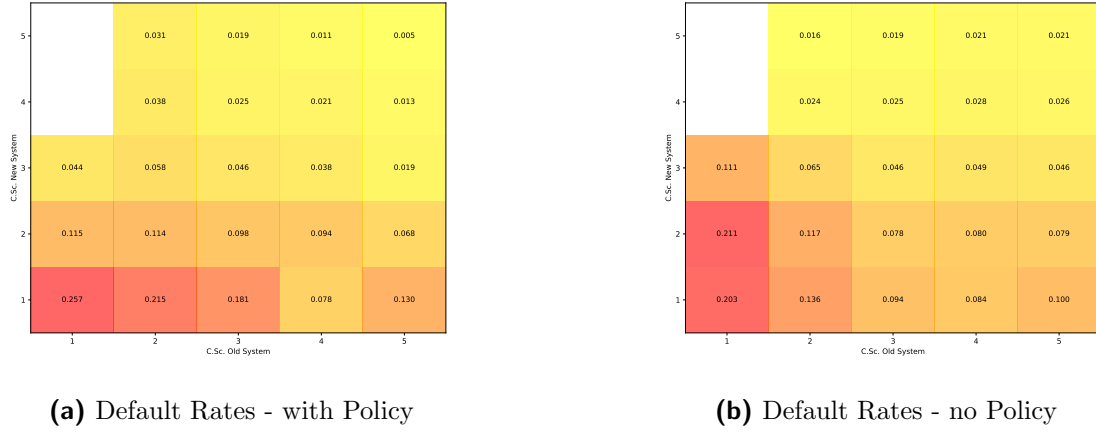
(b) *Horizontal Comparisons*



(c) *Diagonal Comparisons*

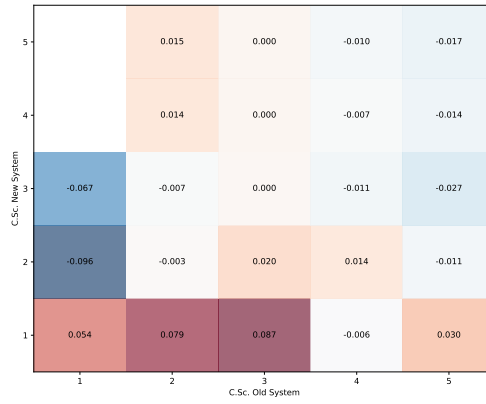
This Figure shows the estimates of coefficients $\{\beta^{kj}\}$ from equation 9 with the same specifications as Panel (c) in Figure 7. The difference is the green lines that correspond to the linear fit between coefficients. A linear fit is computed by regressing values of estimates along each coordinate of the corresponding comparison without weighting. Panel (c) shows the same estimates but rotated to a different angle to better visualize the diagonal comparisons. Table A2 shows the estimates of the linear fits across coordinates.

Figure A12: Estimates of Default Rate



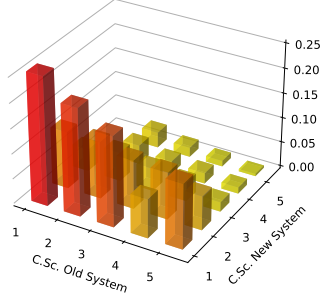
This Figure shows average Default Rates in the periods after the policy. These are the same results from Panels (a) and (b) of Figure 8 in a 2Dimensional perspective. Panel (a) plots our estimates of default rates in the presence of the policy, whereas Panel (b) plots our estimates of Default rates in the absence of the Policy.

Figure A13: Changes in Overall Default Rate

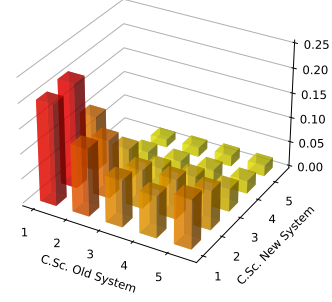


This Figure shows changes in overall default rate for each group. This consists of the difference between average default rate for each group with policy DR_{it}^1 in the periods after the policy and the counterfactual average default rate without the policy in the periods after the policy implementation.

Figure A14: Estimates of Default Rate



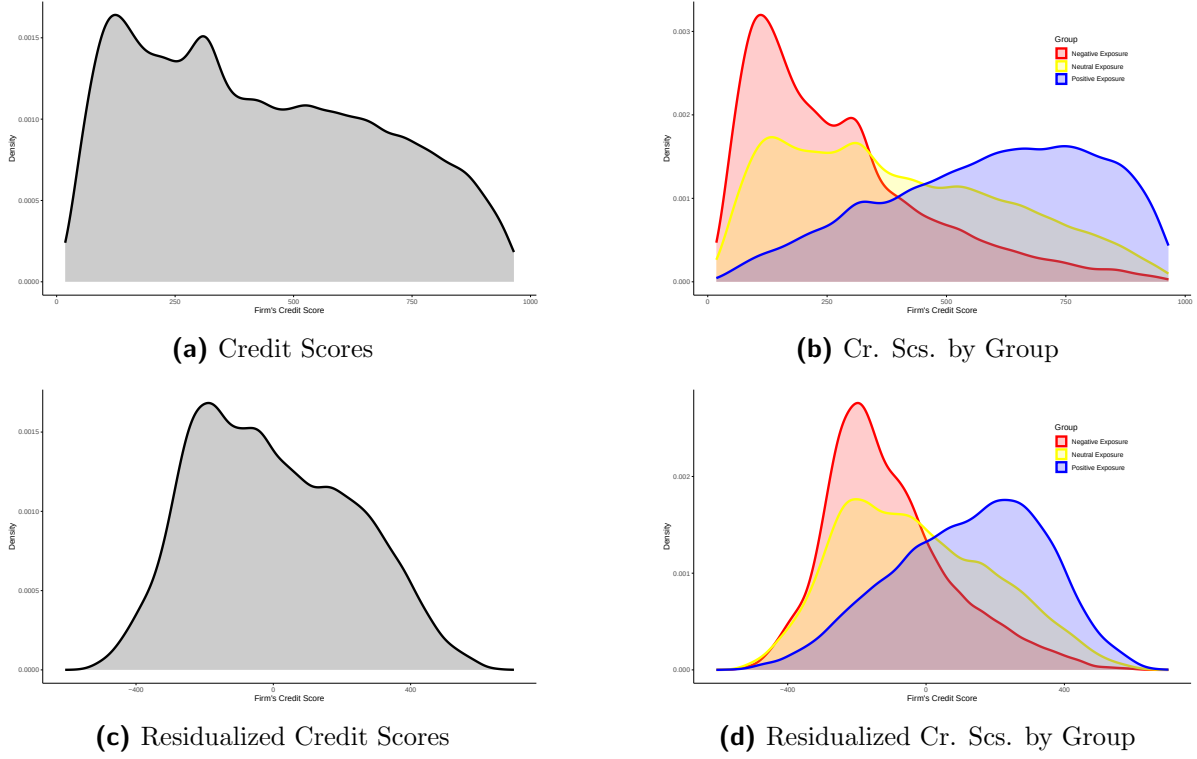
(a) Default Rates - with Policy



(b) Default Rates - no Policy

This Figure shows average Default Rates in the periods after the policy. Panel (a) and (b) are the same results from Figure 8 from a different angle. Panel (a) plots our estimates of default rates in the presence of the policy, whereas Panel (b) plots our estimates of Default rates in the absence of the Policy.

Figure A15: Distribution of Credit Scores of New Firms



This Figure shows k-density estimates of the distribution of credit scores and residualized credit scores calculated in December 2023 for active firms. Residualized credit scores are calculated as the residual of a regression of credit scores on fixed effects of the period of firm entry and cells of demographic characteristics of the entrepreneur, which include age, education, race, and year of birth. Groups of exposure are defined according to the position of the entrepreneur in the joint distribution of credit scores, as described in the body of the paper.

Table A1: Correlation between Credit Scores and Demographic Characteristics

	Negative System	Positive System	Difference in C.Sc.
	(1)	(2)	(3)
Female	-0.059 (0.004)	-0.043 (0.004)	0.016 (0.004)
Nonwhite	-0.170 (0.005)	-0.171 (0.005)	-0.001 (0.004)
High School	0.087 (0.005)	0.226 (0.005)	0.138 (0.005)
College	0.561 (0.007)	0.909 (0.006)	0.349 (0.005)
30	0.186 (0.007)	0.285 (0.008)	0.099 (0.006)
35	0.269 (0.007)	0.477 (0.008)	0.208 (0.006)
40	0.418 (0.007)	0.634 (0.007)	0.216 (0.006)
45	0.356 (0.008)	0.646 (0.008)	0.291 (0.007)
50	0.453 (0.009)	0.729 (0.009)	0.276 (0.007)
55	0.574 (0.010)	0.827 (0.009)	0.253 (0.008)
60	0.780 (0.009)	0.935 (0.008)	0.155 (0.007)
Intercept	-0.475 (0.007)	-0.824 (0.007)	-0.349 (0.006)
Observations	195179	195179	195179
R^2	0.103	0.196	0.036

This Table presents the correlation between Credit Scores and demographic characteristics of individuals. Coefficients are estimates of a linear regression of $\frac{cs_i - \bar{cs}}{sd(cs_i)}$ on observable characteristics. The sample is restricted to the last period before the implementation of the policy. Column (1) shows results using credit scores from the negative system, and Column (2) from the positive system. The outcome in Column (3) is Positive System credit scores minus Negative System credit scores. Robust Standard errors in parenthesis.

Table A2: Linear Fit between Estimates of β^{kj} from Equation 9 with Credit as Outcome

Vertical		Horizontal		Diagonal	
(1)		(2)		(3)	
Fit Between Coordinates		Coordinates		Coordinates	
(1,1) - (1,3)	5791.9119 (1082.2779)	(1,1) - (5,1)	531.8042 (184.4172)	(1,1) - (5,5)	4260.8871 (957.9481)
(2,1) - (2,5)	15049.8305 (2091.4539)	(1,2) - (5,2)	-739.5318 (291.3136)	(4,1) - (5,2)	-1078.6651 (.)
(3,1) - (3,5)	7662.5857 (895.8384)	(1,3) - (5,3)	-1841.8956 (1100.7821)	(3,1) - (5,3)	2072.1172 (367.1220)
(4,1) - (4,5)	5861.3999 (850.7122)	(2,4) - (5,4)	-7952.4951 (2030.8364)	(2,1) - (5,4)	5333.3116 (367.4760)
(5,1) - (5,5)	4198.1683 (1023.4545)	(5,4) - (5,5)	-13132.2642 (3865.9973)	(1,3) - (3,5)	8441.7446 (10781.0828)
				(1,2) - (2,5)	7153.1092 (2188.8593)

This Table shows coefficients and standard errors in parenthesis of linear fit between coefficients β^{kj} from equation 9. Superscript k corresponds to groups of old system of credit scores, and superscript j corresponds to groups of the new system of credit scores. The coordinates $(x_1, y_1) - (x_2, y_2)$ in parenthesis represent the corresponding start and end points of each linear fit. For example, (1,1) - (1,3) corresponds to the linear fit between our estimates of coefficients $\{\beta^{11}, \beta^{12}, \beta^{13}\}$. In turn, (1,1) - (5,1) corresponds to the line between $\{\beta^{11}, \beta^{21}, \beta^{31}, \beta^{41}, \beta^{51}\}$

Table A3: Linear Fit between Estimates of β^{kj} from Equation 9 with Financial Delinquency as outcome

Vertical		Horizontal		Diagonal	
(1)		(2)		(3)	
Fit Between Coordinates		Coordinates		Coordinates	
(1,1) - (1,3)	-1373.5292 (1372.4108)	(1,1) - (5,1)	299.6956 (150.9276)	(1,1) - (5,5)	367.4869 (254.7191)
(2,1) - (2,5)	1633.5986 (302.1776)	(1,2) - (5,2)	1045.7595 (611.7093)	(4,1) - (5,2)	-61.7526 (.)
(3,1) - (3,5)	311.7982 (36.3450)	(1,3) - (5,3)	776.0316 (642.3734)	(3,1) - (5,3)	-129.2677 (40.3109)
(4,1) - (4,5)	142.0436 (57.9121)	(2,4) - (5,4)	-1224.3008 (403.3562)	(2,1) - (5,4)	174.0391 (177.7955)
(5,1) - (5,5)	46.5992 (37.5222)	(5,4) - (5,5)	-1788.8731 (650.1486)	(1,3) - (3,5)	2748.9782 (3170.6027)
				(1,2) - (2,5)	1712.5215 (1133.2098)

This Table shows coefficients and standard errors in parenthesis of linear fit between coefficients γ^{kj} from equation 9 with financial delinquency as the outcome. Superscript k corresponds to groups of old system of credit scores, and superscript j corresponds to groups of the new system of credit scores. The coordinates $(x_1, y_1) - (x_2, y_2)$ in parenthesis represent the corresponding start and end points of each linear fit. For example, (1,1) - (1,3) corresponds to the linear fit between our estimates of coefficients $\{\gamma^{11}, \gamma^{12}, \gamma^{13}\}$. In turn, (1,1) - (5,1) corresponds to the line between $\{\gamma^{11}, \gamma^{21}, \gamma^{31}, \gamma^{41}, \gamma^{51}\}$

Table A4: Comparisons with Additional Sample

	(1)	(2)	(3)
	Credit Sample	Additional Sample	Difference
Female	0.446701	0.458609	0.011908
Nonwhite	0.235102	0.236513	0.001411
Less than H.S.	0.214555	0.201845	-0.012709
High School	0.561083	0.566405	0.005321
Some College	0.224362	0.231750	0.007388
Avg. Age	40.193400	39.146562	-1.046838
Old Sys. C. Sc.	467.149813	473.890127	6.740313
New Sys. C. Sc.	552.889273	548.229299	-4.659974
N. Observations	194235	560172	

This Table shows summary statistics of the sample used for credit analysis and the additional sample included for the entrepreneurship analysis. Credit Score values are from the last period before the implementation of the policy.

Table A5: Comparisons Entrepreneurship Analysis Sample (Risk Set)

	(1)	(2)	(3)
	Excluded	Analysis Sample	Difference
Female	0.455	0.455	0.0005
Nonwhite	0.165	0.248	0.0831
Less than H.S.	0.142	0.215	0.0731
High School	0.558	0.566	0.0080
Some College	0.299	0.218	-0.0812
Age	40.86	39.17	-1.696
Old Sys. C. Sc.	459.61	474.28	14.6
New Sys. C. Sc.	565.50	546.70	-18.79
N. Observations	109367	645040	

This Table shows summary statistics of the sample used in entrepreneurship analysis and the excluded sample. Excluded individuals are those who had already created a firm at any moment before three years of the implementation of the policy. Analysis sample includes the remaining individuals. Credit Score values are from the last period before the implementation of the policy.

Table A6: Firm Quality by Industry

	(1)	(2)	(3)	(4)	(5)
	New Firms		All Firms		
	Num. Emp.	Avg. Wages	Num. Emp.	Total Payroll	Avg. Wages
Positively Exposed	0.055** (0.020)	27.952*** (1.891)	0.836*** (0.129)	3817.944*** (401.742)	46.903*** (2.441)
Negatively Exposed	-0.021 (0.016)	-18.928*** (1.227)	-0.416*** (0.086)	-2038.311*** (241.912)	-32.438*** (1.571)
Omitted group average	3.611	925.095	9.044	20635.467	1083.043
N. Observations	90531	90531	90532	90532	90532

This table shows the characteristics of the industries in which firms created by individuals in different treatment groups operate. It presents point estimates and standard deviations of coefficients β^+, β^- from equation 17. New firm characteristics are constructed using matched employer-employee data three years after creation. "All firms" characteristics are constructed using the average value of outcomes for all firms in the given 5-digit CNAE industry classification. The sample comprises all individuals who created their first firm three years before and two years after the implementation of the policy.

B Data and Sample Construction Details

In this Appendix, we describe our main data sources, go into detail about what each of the main variables is, and describe our main sample construction.

B.1 Data Sources

Our main data source comes from a research partnership with Serasa Experian.

Serasa Experian is the largest credit bureau in Brazil and one of the four authorized by the Central Bank to access the information from Cadastro Positivo. It collects information not only from institutions in the national financial system but also from retail, utility, and insurance companies. In 2007, the company was acquired by Experian, one of the major credit bureaus in the world. It has since been named Serasa Experian, although it is still popularly referred to only as Serasa.

Our data agreement involved major cooperation from the DATALAB in Serasa Experian. That is an area within the company responsible for innovation and research. Their personnel comprises data scientists focused on creating new products from the available data inside the company.

All the data that we, as non-company researchers, accessed was anonymized, and no individual was identifiable. We worked from a virtual computer provided by DATALAB, in which we could access the data and produce our analysis.

Firm Ownership Records (JUCESP): According to the Brazilian constitution, every formal business must register in its state’s trade boards. In Brazil, firm ownership information is in theory public. All states’ Trade Boards are obliged to make all information they collect public but can charge for it.²² JUCESP is the only Trade Board that allows Brazilian Citizens to access information for free. To recover firm records, it is necessary to have a Brazilian Social Security Number and registration on JUCESP’s website. Each citizen has access to 799 firm records per day.

We collect more than 4 million identified firm records, which cover all formal firms created in São Paulo between 2003 and 2023. These records contain all information from the firm ownership history. Importantly, they include the name and social security number of all founders, on top of firm identifiers. This allows us to match firm records from JUCESP with information from the Credit Bureau.

²²Almost all states in the country charge exorbitant price rates for firm records stating who are the owners or who created each firm.

B.2 Main Variables from the Credit Bureau

Our approach attempts to follow as closely as possible the guidelines for using credit agency data defined in [Gibbs et al. \(2024\)](#). Following their classification, we do not access *tradeline-level* data. Instead, our data agreement allowed us to access *tradeline-level* data aggregated to consumer-level. We detail the definition of key variables below.

Credit Scores: we access two credit scores for each individual in our panel. The old system credit scores were constructed using only data sources available to the credit bureau before the policy. In contrast, the new system credit scores represent their updated creditworthiness assessment measure, using data made available by Cadastro Positivo and an updated prediction model. Serasa has multiple credit score measures with specific goals, which they commercialize for different purposes. The ones we use are their most standard measures. Under the new system, the credit bureau defined values above 700 as *excellent* credit scores, those between 501 and 700 as *good*, 301 and 500 as *low*, and below 300 as *very low* credit scores.

Total Credit: is our main outcome variable in the empirical analysis. When creating this variable, we attempted to mimic as closely as possible the definition of *Borrowing* in [Gibbs et al. \(2024\)](#), by summing outstanding balances in all active trade-lines. Total credit is defined at a given period as the sum of outstanding loans and Credit Purchases (mostly credit cards) from a given individual. This includes all types of credit, including credit for consumption and other personal use, and real estate related credit equivalent to mortgages in the US.²³

Financial Delinquency: Financial institutions report the amount owed to credit bureaus when individuals fail to meet their financial obligations on time. Typically, the lender sends a notice indicating the debt and sets a deadline for regularization. If this deadline is not met, the overdue debt may be marked as delinquent and reported to the credit bureau. In our data, each individual’s amount reported as delinquent from financial system obligations is observed at a given period.

Our credit bureau partner receives delinquency measures from other sources besides the financial system. Our data allows us to disentangle between these sources. In our empirical analysis of default rates, we focus on delinquency originating from the financial system. Table 1 shows that delinquency from other sources is, on average, smaller than financial system delinquency.

²³In Brazil, real estate related credit represents a much smaller share of the personal credit market than in the US (around 25% relative to around 70% in the US). Unfortunately, in our data we cannot decompose credit by their type, only by the type of institution. See [Gonzalez et al. \(2023\)](#) for a summary of how personal credit is divided across types in Brazil during our analysis period.

B.3 Sample Construction

We begin with a random sample of 200,000 individuals who live in Sao Paulo. This initial pool of individuals is selected from the universe of formal employees in Sao Paulo between 2015 and 2021, summed with the universe of entrepreneurs (including microentrepreneurs registered as MEI) in the state between 2015 and January 2024. I.e. if an individual had at least one formal job contract in the state of Sao Paulo in the period described above, they are in the pool of potential individuals in the sample. The end product of this step is a list of social security identifiers. In Brazil, the *Cadastro de Pessoa Fisica* (CPF) is the equivalent of Social Security Numbers.

We merge this data with a panel of credit information provided by SERASA. The end product of this step is a panel of individuals with their respective credit profiles.

In the next step, we merge this panel with Entrepreneurship data constructed by scrapping firm records from JUCESP and performing text analysis on them. This data has both the CPF of the entrepreneurs as well as the firm identification number (CNPJ) of the firm they own. We also add gender, age (in bins of five years), education, and race of entrepreneurs, as well as an indicator variable if the firm had multiple owners if the given individual was a founder or joined the company later, and an indicator if the firm is in the MEI tax system.

Due to data construction limitations, we make the entrepreneurship data set uniquely identified at the individual level. This implies that if an individual who owns multiple firms cannot have their CPF-CNPJ pair for all of them. We proceed with the following order of priorities to choose the firm we keep: First, we prioritize non-MEI firms, second we prioritize firms created after 2016, third, we prioritize firms employing someone, then we prioritize if the individual was a founder of that firm, lastly we take the oldest firm among the remaining ones.

C Formalized Conceptual Framework

In this Appendix, we formalize our conceptual framework, with a more precise notation and statement of mathematical assumptions.

C.1 Setup

Borrowers: Potential borrowers differ in their (true) creditworthiness, denoted by $\theta_i \in \mathbb{R}$, with $\theta \sim \mathcal{G}(\cdot)$, where $\mathcal{G}$ is a well-defined density function continuous on $\mathbb{R}$. One could conceptualize this parameter as one governing the profitability of loans and the probability of defaults.

Decision Problem: The lender faces a decision problem. They have an objective function $\pi(C_i, \theta_i)$ where they decide how much credit they supply to an individual (C_i), based on i 's creditworthiness.

A lender does not directly observe the true creditworthiness of the borrower θ_i but receives a signal realization s_i . The information structure can be formally defined as follows. We define a signal S, ρ as a set of Signal realizations $S = \Theta$ and a joint distribution ρ over $\Theta \times S$. The marginal distribution of ρ over Θ is $\mathcal{G}(\cdot)$, and that ρ marginal distribution over S is governed by a distribution $\mathcal{F}(\cdot)$. We assume that a signal realization $s_i \in S$ is an accurate estimate of the true creditworthiness θ_i . That is: $s_i = \mathbb{E}_\rho[\theta_i | s_i]$. The lender then forms expectations of the creditworthiness of the potential borrowers and offers them loans accordingly. The lender's solution to the decision problem is defined as:

$$C^*(s_i) = \operatorname{argmax}_C \mathbb{E}[\pi(C, \theta) | s_i]$$

where the lender maximizes their expected value of the objective function given the signal they receive. To further understand how more informative signals affect credit allocation, we assume that the solution to the decision problem can be written as:

$$C^*(s_i) = g(\mathbb{E}[\theta | s_i])$$

In what follows, we focus on how $\mathbb{E}[\theta | s_i]$ behaves with more informative signals. We then return to our decision problem and discuss what are *rationalizable* changes in credit given a policy that increases the informativeness of signals.

C.2 Gaussian Parametrization

To gain intuition on the effects of the policy, it is useful to include a common parametrization of the signals as a function of the true value of individuals' types. We assume that an individual i 's type θ_i is drawn from an underlying distribution $\theta_i \sim N(\mu, \sigma^2)$. Their signal is a noisy measure of their creditworthiness:

$$s_i = \theta_i + u_i, \quad u_i = \epsilon_i - \Delta_i$$

where Δ_i and ϵ_i are independent and distributed according to $N(0, \sigma_\Delta^2)$ and $N(0, \sigma_\epsilon^2)$ respectively. Thus $u_i \sim N(0, \sigma_\Delta^2 + \sigma_\epsilon^2)$.²⁴ This implies that the lender's expected value about a given borrower's type is then given by:

$$\mathbb{E}[\theta|s_i] = \mu + \underbrace{\frac{\sigma^2}{\sigma^2 + \sigma_\Delta^2 + \sigma_\epsilon^2}}_{\text{Precision Weight}} (s_i - \mu)$$

where the expected value is a linear combination of the prior of the lender, given by the population average, and the signal realization s_i . The *Precision Weight* establishes how much weight the decision maker puts on the signal relative to their prior. It captures how informative the signal is. If it was fully informative ($\sigma_\epsilon^2 + \sigma_\Delta^2 = 0$), the expected value of the individual's type would be equal to the signal. The only difference between the expression above and standard frameworks is that the variance of the *noise* is determined by $\sigma_u^2 = \sigma_\epsilon^2 + \sigma_\Delta^2$.

C.3 Revealing Information

Policy: We now consider a policy that allows Δ_i to be observed. An interpretation of this policy is that signals are constructed with additional information that recovers Δ_i . Thus, lenders observe a new signal $s'_i = s_i + \Delta_i$, which in turn reflects a new expected value of the borrowers' creditworthiness given the signal:

$$\mathbb{E}[\theta|s'_i] = \left(\mu + \frac{\sigma^2}{\sigma^2 + \sigma_\epsilon^2} (s_i + \Delta_i - \mu) \right)$$

To understand the changes in credit given the new signal, we compare the two expected values of creditworthiness, $E[\theta|s'_i]$ and $E[\theta|s_i]$. We can write the difference between them as:

²⁴This is an adaptation of the common formulation of signals in the statistical discrimination literature in labor markets (Phelps, 1972; Aigner and Cain, 1977).

$$\mathbb{E}[\theta|s'_i] - \mathbb{E}[\theta|s_i] = \underbrace{\left(\mu + \frac{\sigma^2}{\sigma^2 + \sigma_\epsilon^2} (s_i + \Delta_i - \mu) \right)}_{\mathbb{E}[\theta_i|s'_i]} - \underbrace{\left(\mu + \frac{\sigma^2}{\sigma^2 + \sigma_\Delta^2 + \sigma_\epsilon^2} (s_i - \mu) \right)}_{\mathbb{E}[\theta_i|s_i]}$$

Define

$$r_1 = \frac{\sigma^2}{\sigma^2 + \sigma_\epsilon^2} \quad r_2 = \frac{\sigma_\Delta^2}{\sigma^2 + \sigma_\epsilon^2 + \sigma_\Delta^2}$$

Then the change in the expected value of θ can be rewritten as:

$$\mathbb{E}[\theta|s'_i] - \mathbb{E}[\theta|s_i] = \underbrace{r_1 \Delta_i}_{\text{Signal Value Change}} + \underbrace{r_1 \cdot r_2 (s_i - \mu)}_{\text{Change in Precision}}$$

The expression above provides us intuition on how an increase in the information changes the expected values of creditworthiness. If the information revealed indicates that the individual is more creditworthy, lenders will update their expectations accordingly. This is straightforward from the *signal value change* part in the expression above

Another implication of our framework is that the revealed information also increases the *precision* of our signal. This implies that expected values of creditworthiness should change even for individuals for whom the information revealed does not change their signal ($\Delta_i = 0$). The intuition behind this is that with a more precise signal, the lenders' precision weight increases; thus, they update the population prior closer to the signal.

We can illustrate this in terms of *indifference curves*. Note that if: $r_1 \Delta + r_1 \cdot r_2 (s_i - \mu) = 0$ then the change in the posterior mean between the new and old signals is zero. This is the set of (s, s') for which $s' = r_2 \cdot \mu + (1 - r_2) \cdot s$. This set includes individuals with $(s_i, s'_i) = (\mu, \mu)$, that is, those with signal equal to the prior and no change with the new information. It also includes all points in the red line in Figure A16.

More generally, all points in a given line:

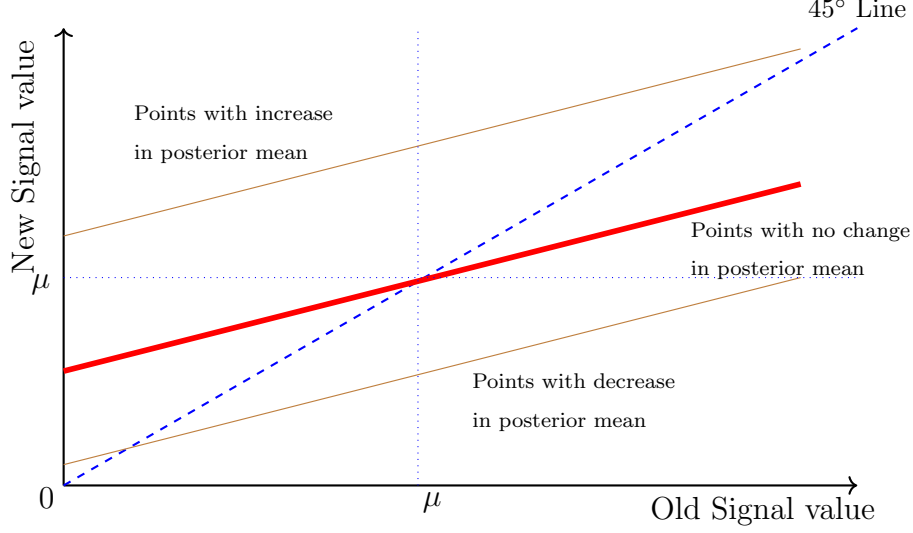
$$s' = k + (1 - r_2) \cdot s$$

have the same change in their posterior mean given by:

$$\mathbb{E}[\theta|s'_i] - \mathbb{E}[\theta|s_i] = r_1 \cdot (k - r_2 \cdot \mu)$$

A key feature of these *indifference curves* is that their slope r_2 will always be positive and smaller than 1. Lines in our diagram with slope equal to 1 (as the 45° Line) represent

Figure A16: Diagram of Indifference Curves in the Changes in Posterior Mean



individuals with the same change in their signal value. As illustrated in Figure A16, our framework implies that conditional on the same signal value change, individuals with higher values of the signal are on higher *indifference curves*, thus implying a higher change in the posterior means.

Generalizing the Gaussian Parametrization:

The Gaussian parametrization of the distributions is useful for exposition purposes but not necessary for our rationalization of changes in credit with more informative signals. In what follows, we state general assumptions on the expected value of creditworthiness given the signal and how the less and more informative signals relate to each other. These add to the information structure described in the framework setup.

Assumption 1: *The expected value of creditworthiness given the signal can be written as a linear combination between the prior and the signal:*

$$\mathbb{E}[\theta|s_i] = \mathbb{E}[\theta] + \frac{1}{a}(s_i - E[\theta])$$

where a is increasing in the variance of the signal realization given the true type.

Assumption 2: *A pre-policy signal s_i is a mean-preserving spread of the new more informative signal s'_i . That implies for a given constant t :*

- (i) $\int_{-\infty}^{\infty} s f(s|\theta) ds = \int_{-\infty}^{\infty} s' f(s'|\theta) ds'$
- (ii) $\int_{-\infty}^t F(s|\theta) ds > \int_{-\infty}^t F(s'|\theta) ds'$

Diaconis and Ylvisaker (1979) establishes sufficiency conditions for distributions to follow the assumption above. In particular, they show that with conjugate priors in the exponential family, it is possible to write the posterior expectation of a random variable with a known distribution given a signal as a convex combination of the expected value and the signal. Changes in the variance of the signal that generate mean-preserving spreads generate changes in the linear combination parameter that go according to the precision weight described above.²⁵

C.4 Implications on Credit Allocation

We now describe how the policy reflects in the lender's solution to their decision problem. In what follows, we assume that the distribution of types and signal realizations follow assumptions 1 and 2.

Recall the solution of the decision problem under the old signal defined as:

$$C^*(s_i) = \operatorname{argmax}_C \mathbb{E}[\pi(C, \theta)|s_i] = g(\mathbb{E}[\theta|s_i])$$

With the more informative signal s' the lender decision problem becomes.

$$C^{**}(s'_i) = \operatorname{argmax}_C \mathbb{E}[\pi(C, \theta)|s'_i]$$

where the expectation is taken over the posterior distribution of θ given the signal s' . We assume:

$$C^{**}(s') = g(\mathbb{E}[\theta|s'])$$

in other words, the credit allocation with the new signal depends only on the posterior mean of θ ; with the same $g(\cdot)$ function as under the old signal.

Under the assumptions above, the changes in the solution of the decision problem for an individual who had an old signal equal to s_i and a new signal s'_i is:

$$h(s_i, s'_i) = C^{**}(s'_i) - C^*(s_i) = g(E[\theta|s'_i]) - g(E[\theta|s_i])$$

We outline propositions that characterize how $h(s_i, s'_i)$ credit should behave over the set comprising values of new and old signals $(s_i, s'_i) \in \mathbb{R}^2$.

²⁵Chambers and Healy (2012) provide more general conditions for what they define as *Update Towards the Signal* types of posteriors. In these cases we still observe a linear relation between the expected value of the random variable and a signal realization. However, they do not explicitly establish how the *precision weight* a relates to the variance of the signal realization.

Proposition 1: *For any given pair of old and new signals, s_i, s'_i , and any given positive constant c , if $g(\cdot)$ is increasing, rationalizable changes in credit should follow:*

- i. $h(s_i, s'_i + c) - h(s_i, s'_i) > 0$
- ii. $h(s_i + c, s'_i) - h(s_i, s'_i) < 0$

Statement (i) and (ii) are what we define as the *Effects of the signal's value*. Their proof is detailed below. (i) states that changes in credit are increasing in the new signal's value. This comes from the fact that, for a fixed value of the old signal s_i , the function that defines changes in credit $h(s_i, s'_i)$ is increasing in s'_i which in turn comes from the solution of the lenders problem being increasing in its argument, which is increasing in s'_i . The same argument is valid for statement (ii). It states that for a given value of the new signal, the changes in credit should be decreasing in the old signal.

The intuition for statement (i) is reasonably straightforward. A signal that implies that individual i is more creditworthy should imply bigger changes in credit. In turn, statement (ii) is slightly less intuitive. We could explain it as for a given value of the new signal, an old signal that implies i is more creditworthy would represent a bigger pre-policy value of credit, which would, in turn, represent a decrease in the credit change.

We can also think about how these statements would fail empirically. If the lender does not perceive the new information available in the new signal as useful, we would expect equality to hold in statement (i). At the same time, if lenders did not perceive information in the old signal as useful, we would expect equality in statement (ii).

Proposition 2: *If $h(s_i, s'_i)$ is increasing in $\mathbb{E}[\theta|s'_i] - \mathbb{E}[\theta|s_i]$, for any given pair of old and new signals, s_i, s'_i , and any given positive constant c , rationalizable changes in credit should follow:*

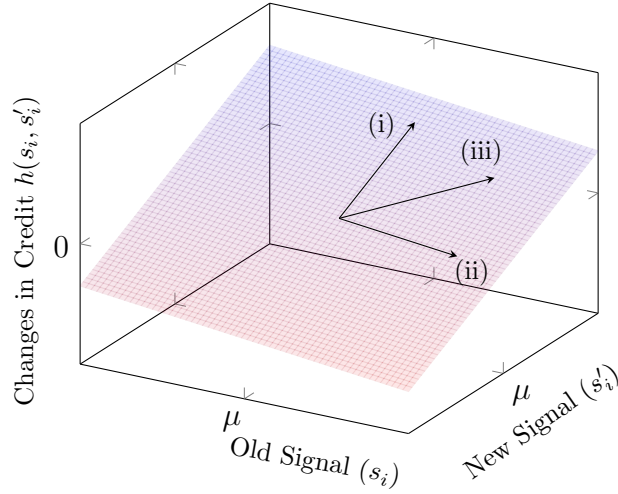
- iii. $h(s_i + c, s'_i + c) - h(s_i, s'_i) > 0$

Statement (iii) defines how conditional on the same difference between signals, changes in credit should behave. If changes in credit are increasing in changes in the posterior of types given a signal, changes in credit should be increasing as we move towards larger values of the signal conditional on the same difference between new and old signal. It requires that we make assumptions on how function $h(s_i, s'_i)$ varies with the differences in $\mathbb{E}[\theta|s'_i] - \mathbb{E}[\theta|s_i]$ instead of assumptions over $g(\cdot)$. It holds because under our assumption over how posteriors are formed given a signal, $(\mathbb{E}[\theta|s'_i + c] - \mathbb{E}[\theta|s_i + c]) > (\mathbb{E}[\theta|s'_i] - \mathbb{E}[\theta|s_i])$. Intuitively, this occurs because, conditional on the same differences in the signal values,

the precision weight multiplied by the signal realization is increasing in the value of the signal. The complete proof is written down in below.

The diagram illustrates how Propositions 1 and 2 allow us to rationalize changes in credit over the joint distribution of signals. It shows changes in credit in the z -axis, values of the old signal (s_i) in the x -axis and values of the new signal (s'_i) in the y -axis. The arrows correspond to each of the statements from both propositions.

Figure A17: Rationalizable Changes in Credit Under a Linear Relationship Between Expected Creditworthiness and Credit



We observe that: (i) conditional on the old signal, changes in credit are increasing in the value of the new signal. (ii) conditional on the new signal, changes in credit are decreasing in the value of the new signal. (iii) conditional on the difference between signals, credit changes are increasing in the value of the signal.

Statement (iii) provides theoretical implications and important consequences of the changes in the credit distribution. As it posits that, conditional on the difference between signals, those with more credit scores should have higher changes, it implies an increase in the inequality in credit access and a widening in the credit distribution, conditional on the changes in credit scores.

When estimating the effects on credit access, we will return to statements (i), (ii), and (iii) to test if our framework can rationalize the changes in credit.

C.5 Propositions Proofs

Proposition: For any given pair of old and new signals, s_i, s'_i , and a given constant c , rationalizable changes in credit should follow:

- i. $h(s_i, s'_i + c) - h(s_i, s'_i) > 0$
- ii. $h(s_i + c, s'_i) - h(s_i, s'_i) < 0$

(i)

$$\begin{aligned}
h(s_i, s'_i + c) - h(s_i, s'_i) &= \left(g(\mathbb{E}[\theta|s'_i + c]) - g(\mathbb{E}[\theta|s_i]) \right) - \left(g(\mathbb{E}[\theta|s'_i]) - g(\mathbb{E}[\theta|s_i]) \right) \\
&= g(\mathbb{E}[\theta|s'_i + c]) - g(\mathbb{E}[\theta|s'_i]) \\
&> 0
\end{aligned}$$

since $\mathbb{E}[\theta|s'_i + c] > \mathbb{E}[\theta|s'_i]$ and $g(\cdot)$ is increasing

(ii)

$$\begin{aligned}
h(s_i + c, s'_i) - h(s_i, s'_i) &= \left(g(\mathbb{E}[\theta|s_i + c]) - g(\mathbb{E}[\theta|s'_i]) \right) - \left(g(\mathbb{E}[\theta|s_i]) - g(\mathbb{E}[\theta|s'_i]) \right) \\
&= g(\mathbb{E}[\theta|s_i + c]) - g(\mathbb{E}[\theta|s_i]) \\
&< 0
\end{aligned}$$

since $\mathbb{E}[\theta|s_i + c] > \mathbb{E}[\theta|s_i]$ and $g(\cdot)$ is increasing.

Proposition 2: *If $h(s_i, s'_i)$ is increasing in $\mathbb{E}[\theta|s'_i] - \mathbb{E}[\theta|s_i]$, for any given pair of old and new signals, s_i, s'_i , and a given positive constant c , rationalizable changes in credit should follow:*

iii. $h(s_i + c, s'_i + c) - h(s_i, s'_i) > 0$

Given the increasing differences assumption, proving statement iii is equivalent to proving that

$$\mathbb{E}[\theta|s'_i + c] - \mathbb{E}[\theta|s_i + c] > \mathbb{E}[\theta|s'_i] - \mathbb{E}[\theta|s_i]$$

We can rewrite this as:

$$\mathbb{E}[\theta|s'_i + c] - \mathbb{E}[\theta|s'_i] > \mathbb{E}[\theta|s_i + c] - \mathbb{E}[\theta|s_i]$$

We can simplify this expression given Assumption 1, which states that $E[\theta|s] = \mathbb{E}[\theta] + \frac{1}{a(s)}(s_i - \mathbb{E}[\theta])$. Here, we slightly change the notation, explaining stating that the term $a(\cdot)$ varies with the signal.

We can rewrite the left and right hand sides of the expression as:

$$\begin{aligned} & \mathbb{E}[\theta] + \frac{1}{a(s')} (s'_i + c - \mathbb{E}[\theta]) - \mathbb{E}[\theta] + \frac{1}{a(s')} (s'_i - \mathbb{E}[\theta]) \frac{1}{a(s')} c \\ & \mathbb{E}[\theta] + \frac{1}{a(s)} (s_i + c - \mathbb{E}[\theta]) - \mathbb{E}[\theta] + \frac{1}{a(s)} (s_i - \mathbb{E}[\theta]) \frac{1}{a(s)} c \end{aligned}$$

From Assumption 2, we know that Variance of s is bigger than the variance of s' , thus from the second part of Assumption 1, we have that $\frac{1}{a(s')} > \frac{1}{a(s)}$.

Therefore:

$$\frac{1}{a(s')} c > \frac{1}{a(s)} c$$

which implies

$$\mathbb{E}[\theta|s'_i + c] - \mathbb{E}[\theta|s'_i] > \mathbb{E}[\theta|s_i + c] - \mathbb{E}[\theta|s_i]$$

D Non Parametric Estimation of $h(s_i, s'_i)$

In addition to the linear projection of changes in credit into the s_i, s'_i presented in Figure 5, we can also estimate $h(s_i, s'_i)$ non-parametrically using sieve-estimators. This is useful as it gives us an additional test of our framework’s empirical propositions and also assesses the quality of the linear fit.

To estimate $h(s_i, s'_i)$ non parametrically we write assume

$$h(s_i, s'_i) \approx h_{np}(s_i, s'_i) = \phi(s_i) + \phi(s'_i)$$

where $\phi()$ are flexible functions of their arguments. We then estimate the following equation using our full sample:

$$Y_{it} = \alpha_i + \delta_t + \phi(\text{C. Sc. Old System}_i) \cdot \text{Post}_t + \phi(\text{C. Sc. New System}_i) \cdot \text{Post}_t + \varepsilon_{it} \quad (\text{D.18})$$

where α_i, δ_t are individual and time fixed effects, and Post_t is a dummy that takes value one for observations after the implementation of the policy. We approximate $\phi()$ using fifth-order polynomials. As in our previous analysis, credit Scores in both old and new systems are normalized measures with mean 0 and standard deviation 1 (Z-scores).

Our coefficient estimates for both linear and non-parametric models are presented in Table A8. But the visualization in Figure A18 is more intuitive. In Panels (a) and (b), we observe the fit of both the nonparametric and linear estimates over the space spanned by s_i, s'_i . In Panel (c), we show how the fit differs between both estimates over the same space.

Visually, the restricted model under the linear assumption is similar to the non-parametric estimation. The latter also predicts that changes in credit are increasing *vertically* and *diagonally*, and decreasing *horizontally* (we further show this below). By looking at Panel (c), we observe that the linear model does a particularly good job in predicting changes around the (0,0) point and a poorer job in the extremities. This is somewhat by construction as $h(0, 0) = 0$ in both models. As discussed, this assumption comes from our conceptual framework. Despite the visual comparison between both estimates suggesting that they are similar, as somewhat expected given the size of our sample, tests of the linear restriction reject that the linear model is sufficient.

On top of the estimated changes in credit, we can also test the implications of our conceptual framework by looking at the derivatives from the estimated shape of the function. Given our estimated coefficients of function $\phi()$ we can recover numerical values for $\frac{\partial h_{np}(s_i, s'_i)}{\partial s_i}$ and $\frac{\partial h_{np}(s_i, s'_i)}{\partial s'_i}$.

We show the values of partial derivatives over the space spanned by (s_i, s'_i) in Figure A19. In Panel (a), we show $\frac{\partial h_{np}(s_i, s'_i)}{\partial s'_i}$, which we label as *vertical* changes consistent with our terms in the paper. Points in which the surface is colored blue indicate positive estimates for the partial derivative, whereas negative values are colored in red. We can observe that partial derivatives are positive in almost every point of the distribution of credit scores, consistent with the Proposition in our conceptual framework.

Our framework also predicts that changes in credit should decrease *horizontally*. We show our estimates for $\frac{\partial h_{np}(s_i, s'_i)}{\partial s_i}$ in Panel (b). We can see that red values dominate in the figure, indicating that in the majority of the distribution, credit changes decrease as we increase the value of the old system credit score.

Figure A19 also allows us to evaluate the last proposition of our framework by observing how changes in credit scores vary *diagonally*. Panel (c) plots values of $\frac{\partial \hat{h}_{np}(s_i, s'_i)}{\partial s'_i} + \frac{\partial \hat{h}_{np}(s_i, s'_i)}{\partial s_i}$, therefore representing a marginal change in the 45-degree lines summarized by $s_i = s'_i + c$. Our estimates suggest that in the majority of the joint distribution of credit scores, $h(s_i, s'_i)$ increases in diagonal comparisons.

Lastly, we can compute the expected value of the derivatives. We do it in two different ways, each with its own interpretation. First, we compute the average of over a grid of equally 1000 points in a space of $s_i \in (-1.5, 1.5) \times s'_i \in (-1.5, 1.5)$. We can define a distribution $\mathcal{F}(s_i, s'_i)$ for this grid with density function $f(s_i, s'_i) = 1/1000$ for points s_i, s'_i in the grid and 0 otherwise. Our average derivatives will then be:

Vertical:

$$\mathbb{E}\left[\frac{\partial \hat{h}_{np}(s_i, s'_i)}{\partial s'_i}\right] = \int \frac{\partial h_{np}(s_i, s'_i)}{\partial s'_i} d\mathcal{F}(s_i, s'_i)$$

Horizontal:

$$\mathbb{E}\left[\frac{\partial \hat{h}_{np}(s_i, s'_i)}{\partial s_i}\right] = \int \frac{\partial h_{np}(s_i, s'_i)}{\partial s_i} d\mathcal{F}(s_i, s'_i)$$

Diagonal:

$$\mathbb{E}\left[\frac{\partial \hat{h}_{np}(s_i, s'_i)}{\partial s'_i} + \frac{\partial \hat{h}_{np}(s_i, s'_i)}{\partial s_i}\right] = \int \frac{\partial h_{np}(s_i, s'_i)}{\partial s'_i} d\mathcal{F}(s_i, s'_i) + \int \frac{\partial h_{np}(s_i, s'_i)}{\partial s_i} d\mathcal{F}(s_i, s'_i)$$

With this method, we input equal weights for all points in the space spanned by s_i, s'_i . Thus, the moment we estimate could be considered an *unweighted* average of partial derivatives.

In another way, we use our sample of individuals with their s_i, s'_i and compute the

average derivatives for each individual using the estimated partial derivatives. We could think of this as a *weighted* average over the joint distribution of signals, with the weights being the empirical distribution of signals. We compute the moments as follows:

Vertical:

$$\mathbb{E}\left[\frac{\partial \hat{h}_{np}(s_i, s'_i)}{\partial s'_i}\right] = \frac{1}{N} \sum_i \frac{\partial h_{np}(s_i, s'_i)}{\partial s'_i}$$

Horizontal:

$$\mathbb{E}\left[\frac{\partial \hat{h}_{np}(s_i, s'_i)}{\partial s_i}\right] = \frac{1}{N} \sum_i \frac{\partial h_{np}(s_i, s'_i)}{\partial s_i}$$

Diagonal:

$$\mathbb{E}\left[\frac{\partial \hat{h}_{np}(s_i, s'_i)}{\partial s_i} + \frac{\partial \hat{h}_{np}(s_i, s'_i)}{\partial s'_i}\right] = \frac{1}{N} \sum_i \frac{\partial h_{np}(s_i, s'_i)}{\partial s_i} + \frac{1}{N} \sum_i \frac{\partial h_{np}(s_i, s'_i)}{\partial s'_i}$$

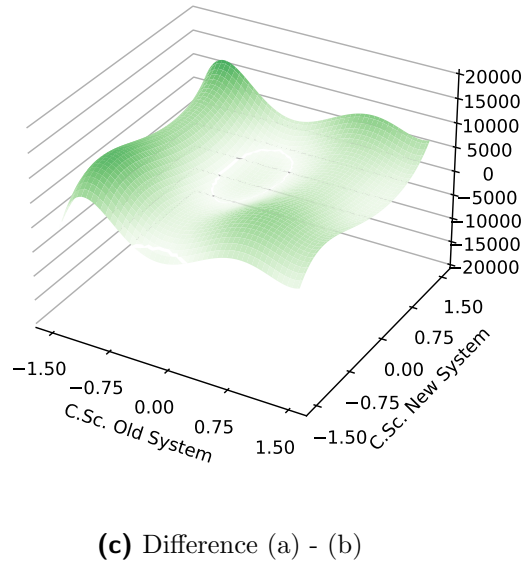
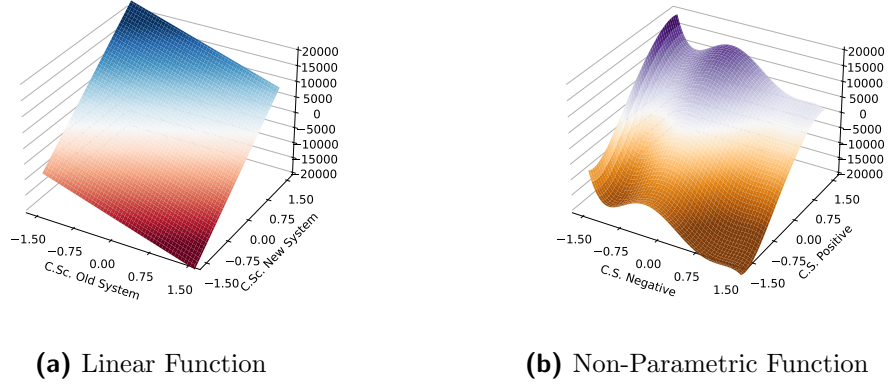
Our estimates for both methods suggest that on the three *directions*, the non-parametric model goes in line with our conceptual framework predictions. Changes in credit increase with respect to the value of the new signal and decrease with respect to the old signal. Furthermore, *vertical* increases are larger than *horizontal* ones, thus indicating that changes in credit also increase *diagonally*.

Table A7: Expected Values of Partial Derivatives in the Non-Parametric Estimation

		(1)	(2)	
Over the Grid			Over the Sample	
	Partial Derivative	Value	Partial Derivative	Value
<i>Vertical</i>	$\int \frac{\partial \hat{h}_{np}(s_i, s'_i)}{\partial s'_i} d\mathcal{F}(s_i, s'_i)$	9137.39	$\frac{1}{N} \sum_i \frac{\partial \hat{h}_{np}(s_i, s'_i)}{\partial s'_i}$	6731.50
<i>Horizontal</i>	$\int \frac{\partial \hat{h}_{np}(s_i, s'_i)}{\partial s_i} d\mathcal{F}(s_i, s'_i)$	-6104.38	$\frac{1}{N} \sum_i \frac{\partial \hat{h}_{np}(s_i, s'_i)}{\partial s_i}$	-6278.01
<i>Diagonal</i>	$\int \frac{\partial \hat{h}_{np}(s_i, s'_i)}{\partial s'_i} d\mathcal{F}(s_i, s'_i) + \int \frac{\partial \hat{h}_{np}(s_i, s'_i)}{\partial s_i} d\mathcal{F}(s_i, s'_i)$	3033.01	$\frac{1}{N} \sum_i \frac{\partial \hat{h}_{np}(s_i, s'_i)}{\partial s'_i} + \frac{1}{N} \sum_i \frac{\partial \hat{h}_{np}(s_i, s'_i)}{\partial s_i}$	453.49

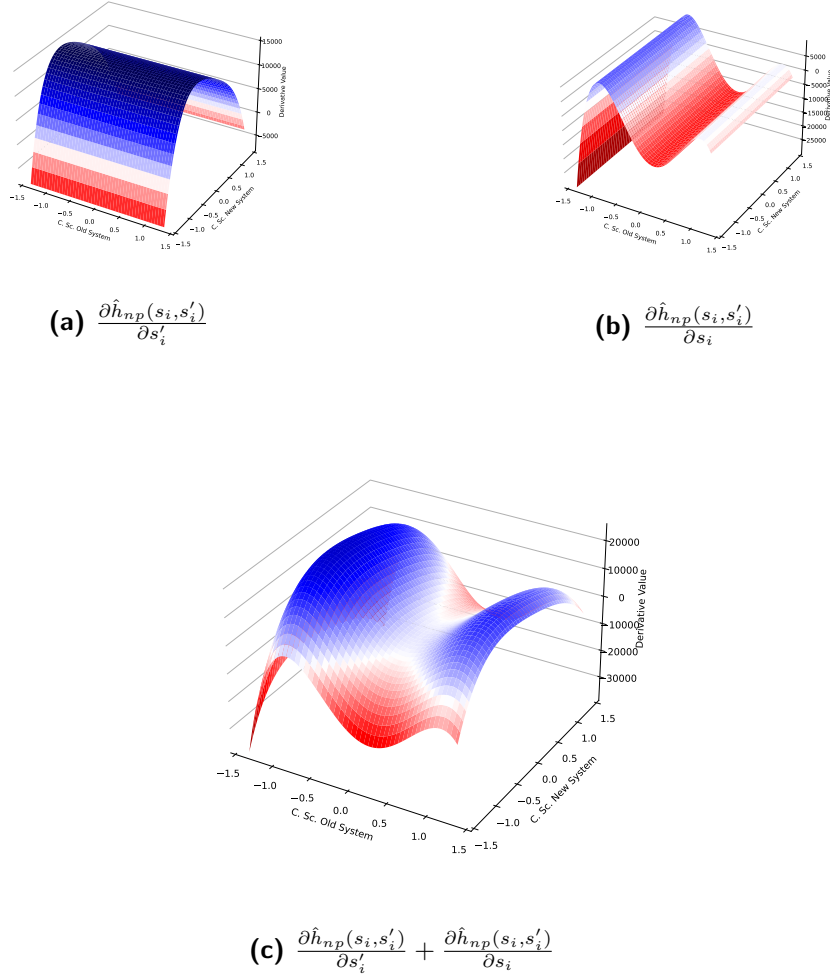
In this Table we present the estimated values of partial derivatives using non parametric estimation. *Over the grid* estimates assume a 1000 point grid over $s_i \in (-1.5, 1.5) \times s'_i \in (-1.5, 1.5)$ with equal weights to each point. In turn, *Over the Sample* calculates partial derivatives to every individual in our sample and weights each individual equally.

Figure A18: Comparisons Between Linear and Non-Parametric Estimates of Credit Change



Panel (a) and (b) show predicted values of changes in credit over the space spanned by old and new system credit scores using our linear restricted model (equation 10) and the non-parametric model (equation D.18), respectively. Panel (c) shows the differences between estimates from (a) and (b).

Figure A19: Comparisons Between Linear and Non-Parametric Estimates of Credit Change



This Figure plots partial derivatives calculated over the joint distribution of credit scores in the old and new systems. We use our estimates from equation D.18 and calculate partial derivatives over a grid of 100 equally distributed points over the space spanned by s_i, s'_i

Table A8: Coefficients of Linear and Nonlinear Estimates

	(1)	(2)
	Dependent variable: Credit	
	Linear	Polynomials
C. Sc. Old System	-5172.3*** (273.66)	-11026.4486*** (1471.2169)
C. Sc. Old System ²		-9494.6893*** (1087.5093)
C. Sc. Old System ³		11295.7751*** (2009.2653)
C. Sc. Old System ⁴		3954.4166*** (507.6831)
C. Sc. Old System ⁵		-4041.9399*** (651.5566)
C. Sc. New System	10951.58*** (370.70)	14178.6652*** (721.0755)
C. Sc. New System ²		-2217.2853*** (732.6252)
C. Sc. New System ³		-1089.4464** (532.9137)
C. Sc. New System ⁴		636.4236*** (226.3324)
C. Sc. New System ⁵		-508.6011*** (184.7629)
Observations	2875942	2875942

This Table presents coefficient estimates of equations 10 and D.18. Credit Scores in both old and new systems are normalized measures with a mean of 0 and a standard deviation of 1 (Z-scores). Standard errors in parenthesis are clustered at the individual level.

E Heterogeneity Estimates Across Demographic Groups

In this Appendix we formalize our empirical strategy and provide the full set of estimates to characterize how the policy affected individuals differently according to their

demographic characteristics.

Let $G_i \in \{0, 1\}$ represent a binary variable that defines a specific demographic group, such as male vs. female or white vs. nonwhite. Additionally, let C_{i0} and C_{i1} represent individual credit access before and after the implementation of the policy, respectively.

We are interested in comparing the effect of the policy on credit access across different demographic groups. Specifically, we analyze the difference in the expected change in credit access between two groups. Formally, we make comparisons of the following type:

$$\mathbb{E}[C_{i1} - C_{i0}|G_i = 1] - \mathbb{E}[C_{i1} - C_{i0}|G_i = 0].$$

To estimate the differences in credit access before and after the policy for each group, we rely on our semi-parametric approach, exploring the 5 groups of the old credit score $\mathcal{K} \in \{1, 2, 3, 4, 5\}$ and the 5 groups of the new system credit scores $\mathcal{J} \in \{1, 2, 3, 4, 5\}$. This generates the following estimating equation:

$$Y_{it} = \alpha_i + \delta_t + \sum_{k \in \mathcal{K}} \sum_{j \in \mathcal{J}} \gamma_{g0}^{kj} \cdot D_i^k \cdot D_i^j \cdot \text{Post}_t + \sum_{k \in \mathcal{K}} \sum_{j \in \mathcal{J}} \gamma_{g1}^{kj} \cdot D_i^k \cdot D_i^j \cdot \text{Post}_t \cdot G_i + \varepsilon_{it}. \quad (\text{E.19})$$

Here, α_i represents individual-specific effects, δ_t represents time-specific effects, D_i^k and D_i^j represent credit score categories, and Post_t is a dummy variable indicating the post-policy period. The coefficients γ_{g0}^{kj} capture the impact of the policy for the baseline group, while the interaction terms $\gamma_{g1}^{kj} \cdot G_i$ capture the differential effect for group $G_i = 1$.

Given these estimates, we define the expected change in credit access for each group as follows:

$$\begin{aligned} \mathbb{E}[C_{i1} - C_{i0}|G_i = 0, D_i^k = 1, D_i^j = 1] &= \hat{\gamma}_{g0}^{kj} \\ \mathbb{E}[C_{i1} - C_{i0}|G_i = 1, D_i^k = 1, D_i^j = 1] &= \hat{\gamma}_{g0}^{kj} + \hat{\gamma}_{g1}^{kj}. \end{aligned}$$

This allows us to express the expected change in credit access for each group as follows:

$$\begin{aligned} \mathbb{E}[C_{i1} - C_{i0}|G_i = 0] &= \mathbb{E}[\hat{\gamma}_{g0}^{kj}] \\ \mathbb{E}[C_{i1} - C_{i0}|G_i = 1] &= \mathbb{E}[\hat{\gamma}_{g0}^{kj} + \hat{\gamma}_{g1}^{kj}]. \end{aligned}$$

To calculate the value of expected changes by group we weight the coefficient change by the amount of people of the given group in each of the points of our grid that divides the joint distribution of credit scores. This is equivalent to:

$$\mathbb{E}[C_{i1} - C_{i0}|G_i = 0] = \mathbb{E}[\hat{\gamma}_{g0}^{kj}] = \frac{\sum_i (1 - G_i) \cdot D_i^k \cdot D_i^j \hat{\gamma}_{g0}^{kj}}{\sum_i (1 - G_i) \cdot D_i^k \cdot D_i^j}$$

$$\mathbb{E}[C_{i1} - C_{i0}|G_i = 1] = \mathbb{E}[\hat{\gamma}_{g0}^{kj} + \hat{\gamma}_{g1}^{kj}] = \frac{\sum_i G_i \cdot D_i^k \cdot D_i^j (\hat{\gamma}_{g0}^{kj} + \hat{\gamma}_{g1}^{kj})}{\sum_i G_i \cdot D_i^k \cdot D_i^j}$$

Thus, we can write the difference in policy effects between the two groups as:

$$\begin{aligned} \mathbb{E}[C_{i1} - C_{i0}|G_i = 1] - \mathbb{E}[C_{i1} - C_{i0}|G_i = 0] &= \frac{\sum_i G_i \cdot D_i^k \cdot D_i^j (\hat{\gamma}_{g0}^{kj} + \hat{\gamma}_{g1}^{kj})}{\sum_i G_i \cdot D_i^k \cdot D_i^j} \\ &\quad - \frac{\sum_i (1 - G_i) \cdot D_i^k \cdot D_i^j \hat{\gamma}_{g0}^{kj}}{\sum_i (1 - G_i) \cdot D_i^k \cdot D_i^j} \end{aligned} \quad (\text{E.20})$$

The expression suggests that to understand which group is more affected by the policy, we must consider two components. First, the groups are potentially differently distributed across the joint distribution of credit scores. Second, the heterogeneous treatment effects, captured by the values of $\hat{\gamma}_{g1}^{kj}, \hat{\gamma}_{g0}^{kj}$.

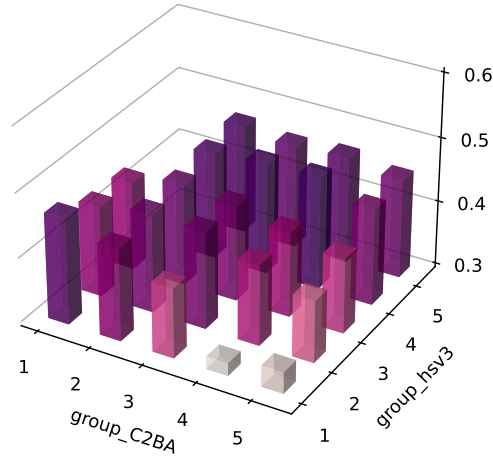
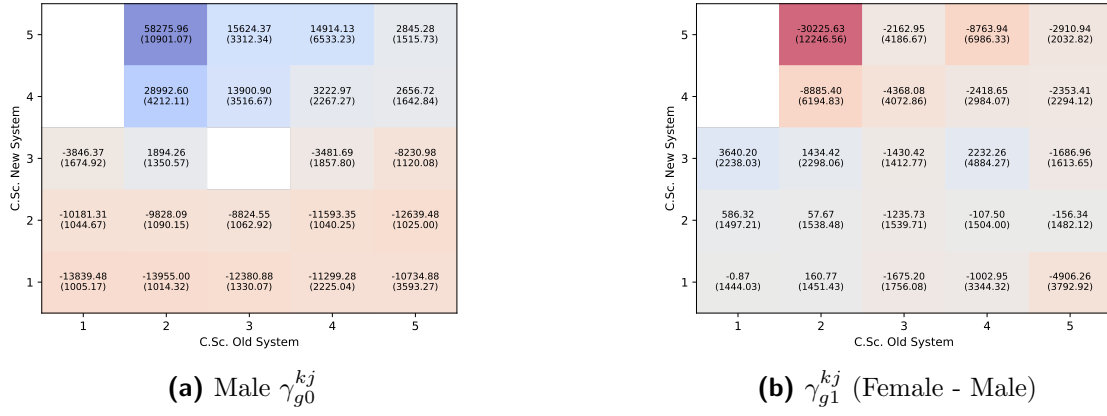
To disentangle between both effects, we can also make an Oaxaca-Blinder type decomposition, replacing the coefficients above by $\hat{\gamma}^{kj}$ from equation 9. This would give us just the effects of the differential composition of individuals by group. This essentially means calculating the following formula

$$\frac{\sum_i G_i \cdot D_i^k \cdot D_i^j (\hat{\gamma}^{kj})}{\sum_i G_i \cdot D_i^k \cdot D_i^j} - \frac{\sum_i (1 - G_i) \cdot D_i^k \cdot D_i^j \hat{\gamma}^{kj}}{\sum_i (1 - G_i) \cdot D_i^k \cdot D_i^j}$$

where $\hat{\gamma}^{kj}$ are the coefficients estimated from equation 9 and presented in Figure 4

We focus our analysis on comparisons across four key demographic dimensions: gender, race, education, and age. We present our full estimates below. These are used to construct table 2 according to the procedure described above.

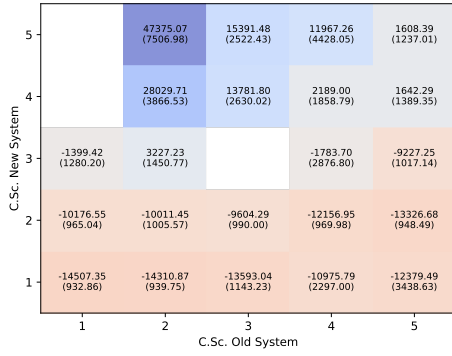
Figure A20: Gender by Group of the Joint Distribution of Credit Scores



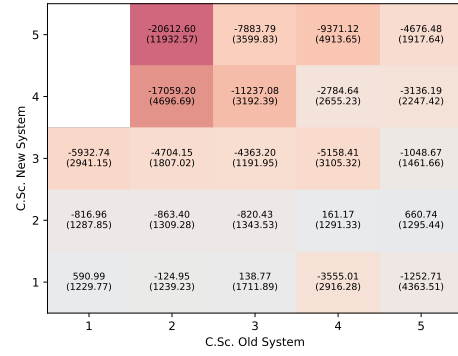
(c) Share of Women across the Joint Distribution of Credit Scores

In Panels (a) and (b), we plot our coefficient estimates from Equation 11. $G_i = 1$ indicates that individual i is a woman. These are the values used to calculate our expression in equation 12. Standard errors are shown in parenthesis. In Panel (c) we show the share of women across the joint distribution of credit scores.

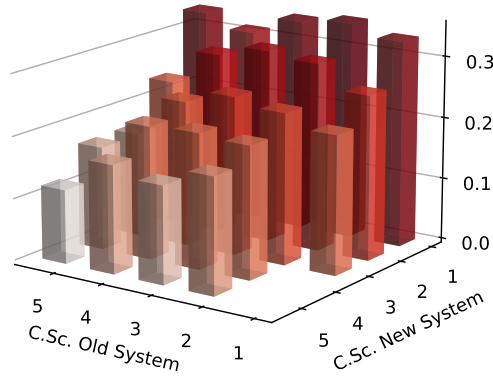
Figure A21: Changes in Credit by Race by Group of the Joint Distribution of Credit Scores



(a) White γ_{g0}^{kj}



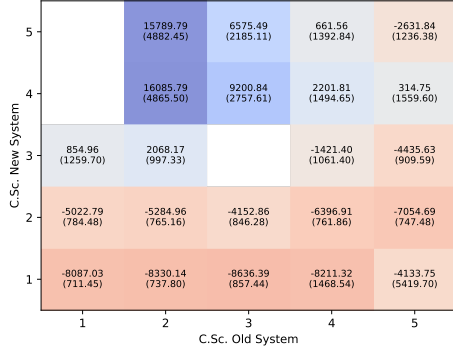
(b) γ_{g1}^{kj} (Nonwhite - White)



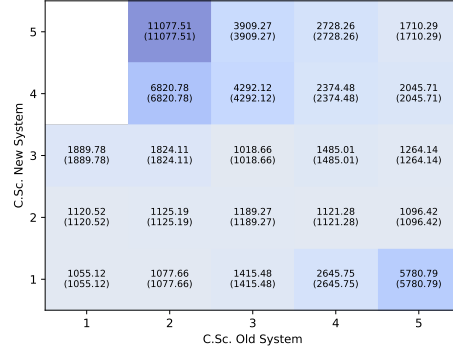
(c) Share of Nonwhite by Group

In Panels (a) and (b), we plot our coefficient estimates from Equation 11. $G_i = 1$ indicates that individual i is a nonwhite individual. These are the values used to calculate our expression in equation 12. Standard errors are shown in parentheses. In Panel (c), we show the share of nonwhite individuals across the joint distribution of credit scores.

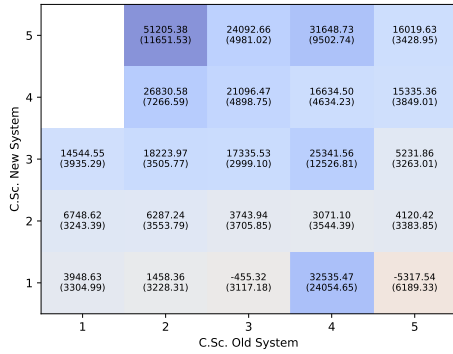
Figure A22: Changes in Credit by Education by Group of the Joint Distribution of Credit Scores



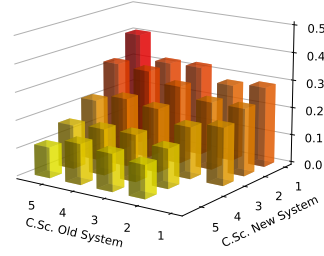
(a) Less than H.S. γ_{g0}^{kj}



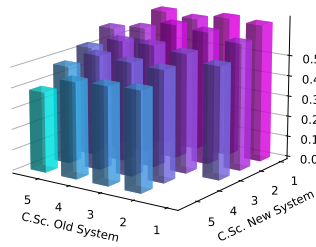
(b) γ_{g1}^{kj} (H.S. - < H.S.)



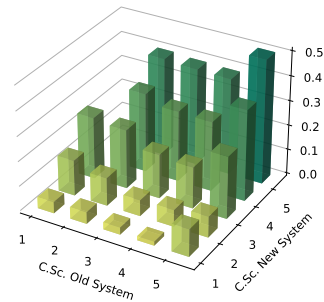
(c) γ_{g2}^{kj} (Coll. - < H.S.)



(d) Share with Less than H.S.



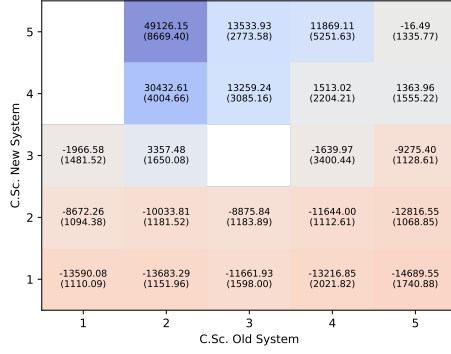
(e) Share with High School



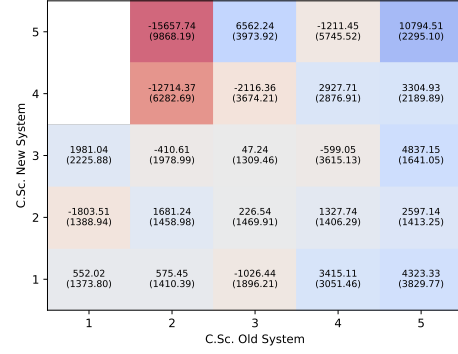
(f) Share with Some College

Figure A23: In Panels (a) and (b), we plot our coefficient estimates from Equation 11. We add a second heterogeneity group, leaving at base individuals with less than high school. These are the values used to calculate our expression in equation 12. Standard errors are shown in parentheses. In Panel (c), we show the share of individuals with each educational attainment across the joint distribution of credit scores.

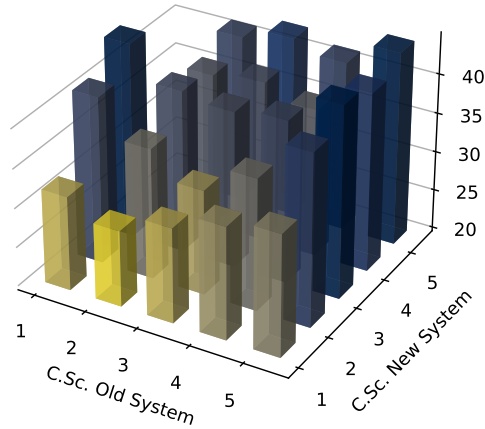
Figure A24: Changes in Credit by Age Group by Group of the Joint Distribution of Credit Scores



(a) 40 y.o. or more γ_{g0}^{kj}



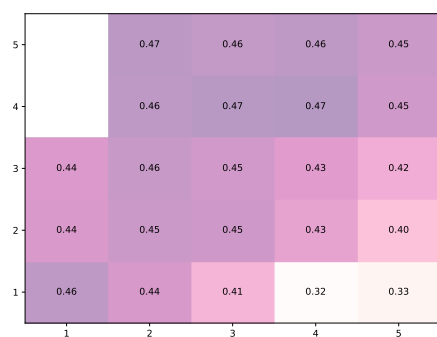
(b) γ_{g1}^{kj} (Below 40 - 40 y.o. or more)



(c) Avg. Age by Group

In Panels (a) and (b), we plot our coefficient estimates from Equation 11. $G_i = 1$ indicates that individual i is less than 40 years old. These are the values used to calculate our expression in equation 12. Standard errors are shown in parenthesis. In Panel (c) we show the average age of individuals across the joint distribution of credit scores.

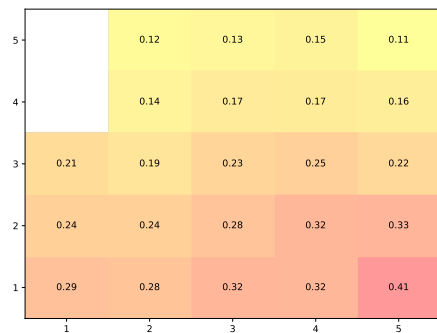
Figure A25: Heatmap by Group of the Joint Distribution of Credit Scores



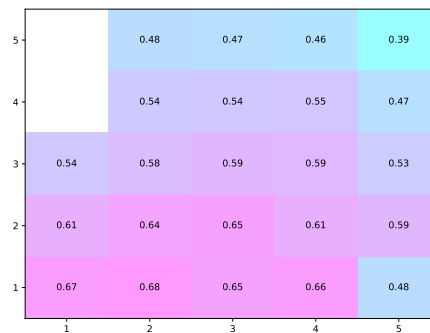
(a) Share Women



(b) Share Nonwhite



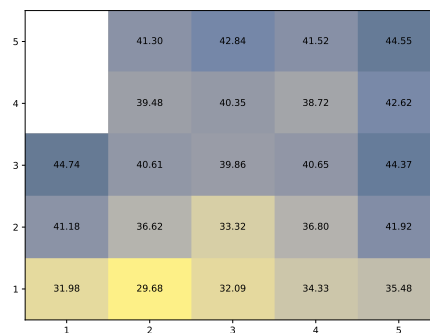
(c) Share with Less than H.S.



(d) Share with High School



(e) Share with Some College



(f) Avg. Age

This Figure shows heatmap plots of how demographic characteristics vary across the joint distribution of credit scores. These are the same results from Panel (c) in Figures A20,A21,A24 and Panels (d), (e), (f) of Figure A22.

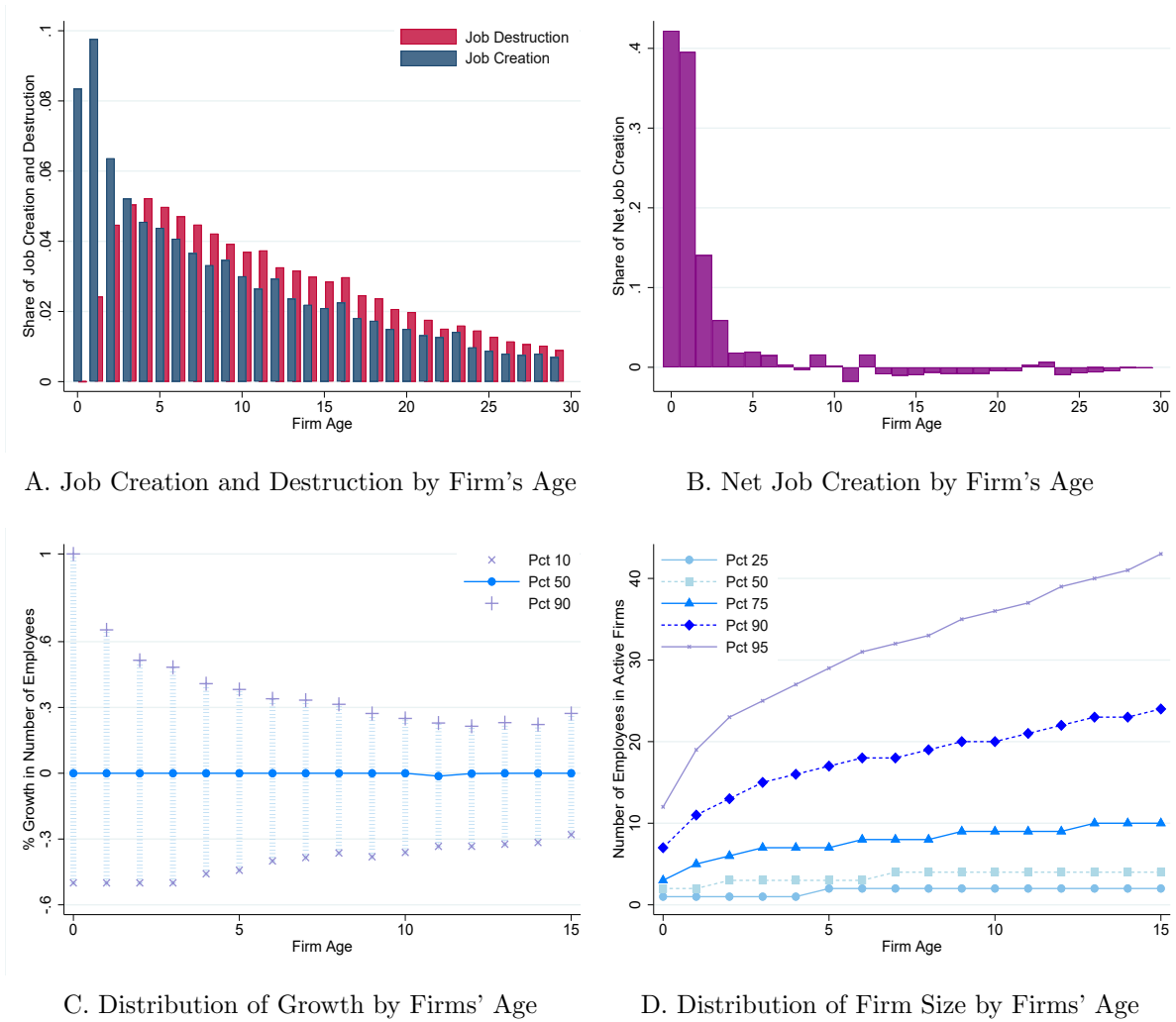
F Appendix - Entrepreneurs and Job Creation in Sao Paulo

Startups in São Paulo mimic behavior that has been documented in the United States. They are responsible for a large share of job creation in the economy. Nevertheless, most startups start small and remain small through time, conditional on survival. In this section, we replicate patterns documented for the U.S. in [Decker et al. \(2014\)](#) combining our entrepreneurship records from JUCESP with matched employer-employee data that covers the universe of formal labor markets (RAIS).

These patterns are illustrated in Figure [A26](#), which are constructed using firm-level data from 2003-2015. We use an older period for this analysis as by the time this paper is being written, more recent RAIS data that allows us to follow firms' for at least 5 years after their creation is not yet available.

Panel A and B reflect job creation and destruction. A job is considered to be created if, for a given firm, one employee was added between year t and year $t+1$. The opposite holds for job destruction. Panel A shows that firms in their first 5 years of existence are responsible for around 35% of all jobs created in the period. In Panel B, we see that net job creation, calculated as job creation minus job destruction, is especially high in the first two years of a firm's history. This is found mechanically, as firms that have just been opened have no job destruction. In Panel C, we see that the median growth in the number of employees is 0 throughout all the first 15 years of firms' existence. Furthermore, we see that in the firms' first years, there is a higher variance in firm growth, which stabilizes around year 6. Lastly, in Panel D, we observe that firms are usually small in number of employees. The median startup in Sao Paulo starts with 3 employees, and by year 10, among active firms, the median firm still only employs 5 people.

Figure A26: Startups Job Creation and Growth



This figure shows statistics for job creation and firm growth for all firms created in the state of São Paulo between 2003 and 2015.

G Informality

In our main entrepreneurship analysis, our focus lies on formal firms. This is due to the fact that both our firm ownership records and the credit bureau data comprise only information from firms with a registration number in the *Cadastro Nacional de Pessoas Jurídicas* (CNPJ). In this Appendix, we discuss the role of informality in entrepreneurship, providing some descriptive evidence using the National Household Survey (PNAD-C) that encompasses both the formal and informal sectors.

First, we describe our informality definition. Brazil is a setting where this is very clear. For employees, all formal labor relations must have a "signed" *carteira de trabalho*.²⁶ Although this dichotomous distinction masks informal labor ties between formal workers (see [Feinmann et al. \(2022\)](#) for a full discussion about this issue), it is still useful as it is observable in the household survey. Informal employees represent 15% of all employees in the state of São Paulo and 13% of all non-domestic employees.

In the case of firms, we use as an indicator of formality the registration in the *Cadastro Nacional de Pessoas Jurídicas*. is the Brazilian National Registry of Legal Entities. It is a unique identifier issued by the Brazilian Federal Revenue Service (Receita Federal) to businesses and other legal entities operating in Brazil. The CNPJ number is similar to a tax identification number in other countries and is required for businesses to conduct legal and financial activities within the country. In PNAD-C, self-employed individuals and employers are asked if their business is registered in CNPJ.

Given the definition of formal businesses, we can describe the differences between formal and informal entrepreneurs in this setting. Given the PNAD-C questionnaire, we have to use a broad definition of entrepreneurs, which encompasses employers and self-employed individuals without employees. In our sample, 25% of entrepreneurs are employers.

We summarize the informality levels of entrepreneurs in Table [A9](#). On average, 90.8% of employers' businesses are registered. In turn, 38.5% of self-employed individuals have a registration in CNPJ. When summing employers and self-employed, 51.4% of entrepreneurs have their businesses registered.

We further describe the informality rates of self-employed by economic sectors in Table [A10](#). In Columns (1) and (2) we show the share of self-employed and employers in each sector. In Columns (3) and (4) we show the share of formal businesses in each sector. We observe that construction sector has the lowest formalization rates both

²⁶In the past, this consisted of an actual booklet for each employee, in which employers had to sign and register contract wages and hours. Nowadays, the process can be done fully electronically. Nonetheless, Brazilians still refer to formal job ties as those with "signed" *carteira de trabalho*.

Table A9: Informality among Entrepreneurs

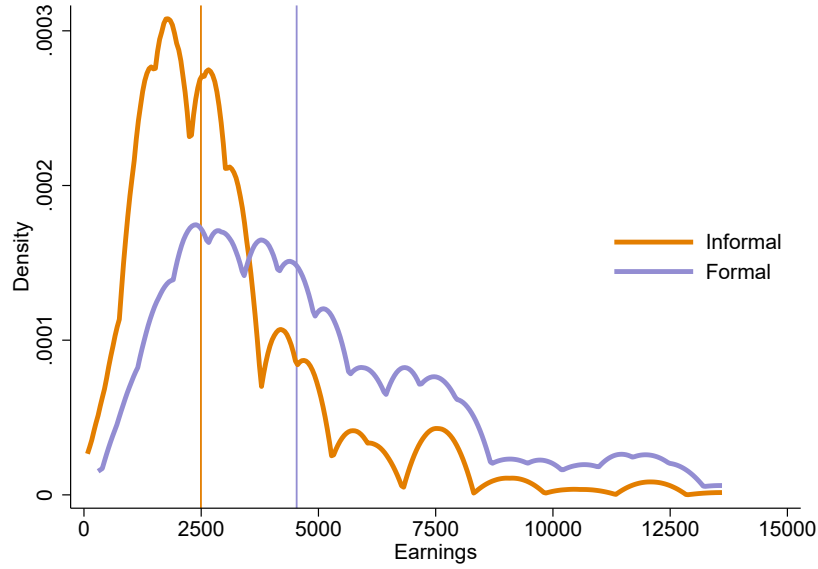
	(1)	(2)
	Informal	Formal
Self-Employed	61.51	38.49
Employers	9.16	90.84
Entrepreneurs	48.63	51.37

The sample is restricted to those between 18 and 65 years old in the State of São Paulo working at least 30 hours a week as entrepreneurs in their main job. Employers are those who self-report as *empregadores*, and self-employed are those who report as *conta-própria*. Entrepreneurs encompass both groups. Formal is defined as if their business had a registration in the CNPJ.

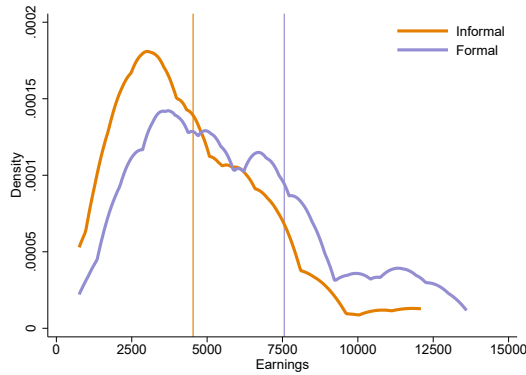
among employers and self-employed individuals.

Lastly, we characterize the earnings distributions of formal and informal entrepreneurs in Figure A27. We observe that the earnings of informal entrepreneurs are substantially lower than those of formal ones. This is valid for both employers and employees. The median earnings of formal entrepreneurs are almost double that of informal ones.

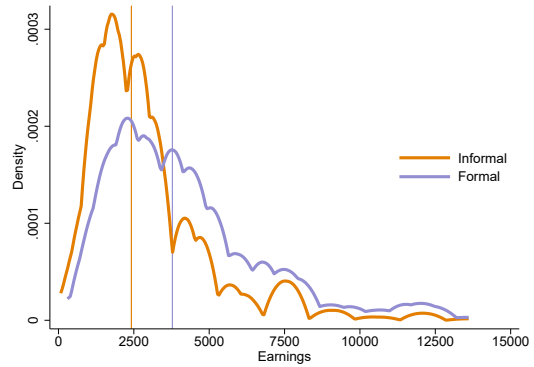
Figure A27: Entrepreneurs Earnings in the Formal and Informal Sector



(a) Entrepreneurs



(b) Employers



(c) Self-Employed

This Figure shows the distribution of labor earnings of entrepreneurs in the State of São Paulo in the last quarter of 2019 built with PNAD-C. The sample is restricted to those between 18 and 65 years old working at least 30 hours a week as entrepreneurs in their main job. Values are adjusted for December 2023 BRL. Employers are those who self-report as *empregadores*, and self-employed are those who report as *conta-própria*. Entrepreneurs encompass both groups. Formal is defined as if their business had a registration in the CNPJ. Vertical lines correspond to median earnings of each group.

Table A10: Share of Business Registration by Economic Sectors

	(1)	(2)	(3)	(4)
	Share in each Sector		Share of Formal Businesses	
	Self-Employed	Employers	Self-Employed	Employers
Agriculture	3.56	4.32	57.32	85.21
Manufacturing	8.61	9.45	32.41	92.39
Construction	17.61	6.59	20.40	71.47
Retail	20.19	31.39	51.62	97.38
Transport & Food	19.88	17.06	28.96	88.10
Professional Services	17.64	25.23	51.84	90.90
Domestic Services	12.51	5.97	38.11	87.05

The sample is restricted to those between 18 and 65 years old in the State of São Paulo working at least 30 hours a week as entrepreneurs in their main job. Employers are those who self-report as *empregadores*, and self-employed are those who report as *conta-própria*. Entrepreneurs encompass both groups. Formal is defined as if their business had a registration in the CNPJ. Sectors are constructed using the first 2-digit of cnae as follows values 1-9 (Agriculture), 2 for 10-39 (Manufacturing), 3 for 41-43 (Construction), 4 for 45-48 (Retail), 5 for 49-56 (Transport & Food), 6 for 58-88 (Professional Services), and 7 for 90-99 (Domestic Services).